

## Chapter 75

# Preferences

This chapter covers the various options that are available from the Fusion Preferences Window.

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# Preferences Overview

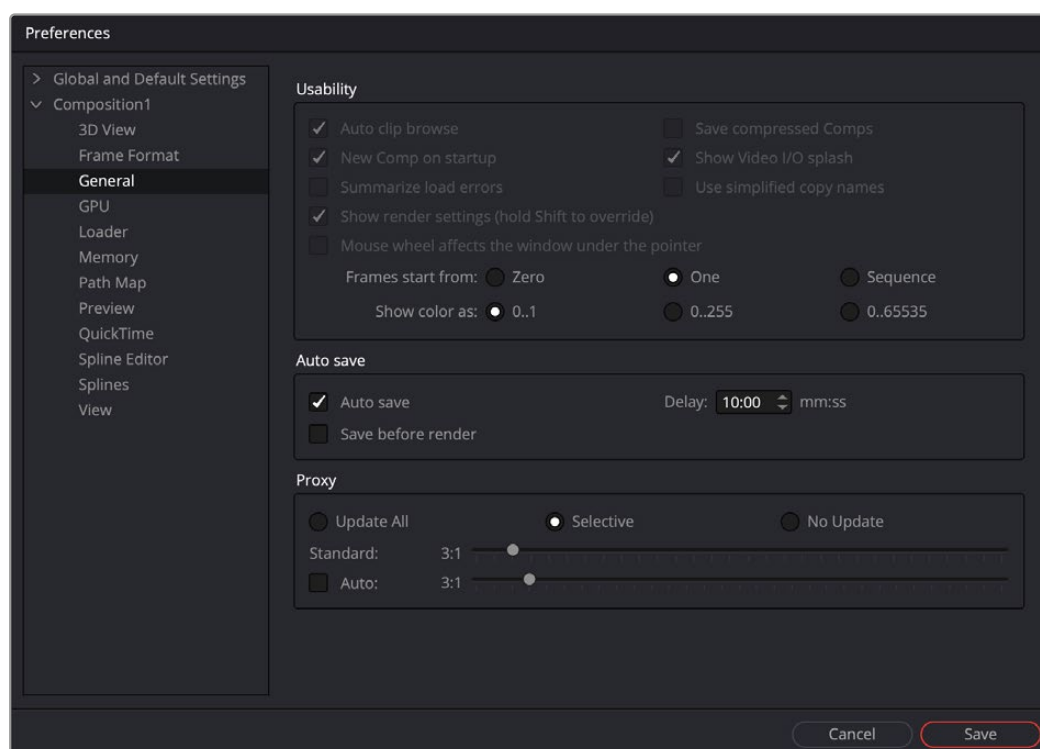
The Preferences window provides a wide variety of optional settings available for you to configure Fusion's behavior to better suit your working environment. These settings are accessed via the Preferences window. The Preferences window can be opened from a menu at the top of the interface.

**In DaVinci Resolve, to open the Fusion Preferences window, do one of the following:**

- On macOS, switch to the Fusion page and choose Fusion > Fusion Settings.
- On Windows, switch to the Fusion page and choose Fusion > Fusion Settings.
- On Linux, switch to the Fusion page and choose Fusion > Fusion Settings.

**In Fusion Studio, to open the Fusion Preferences window, do one of the following:**

- On macOS, choose Fusion Studio > Preferences.
- On Windows, choose File > Preferences.
- On Linux, choose File > Preferences.



The Preferences window includes composition settings.

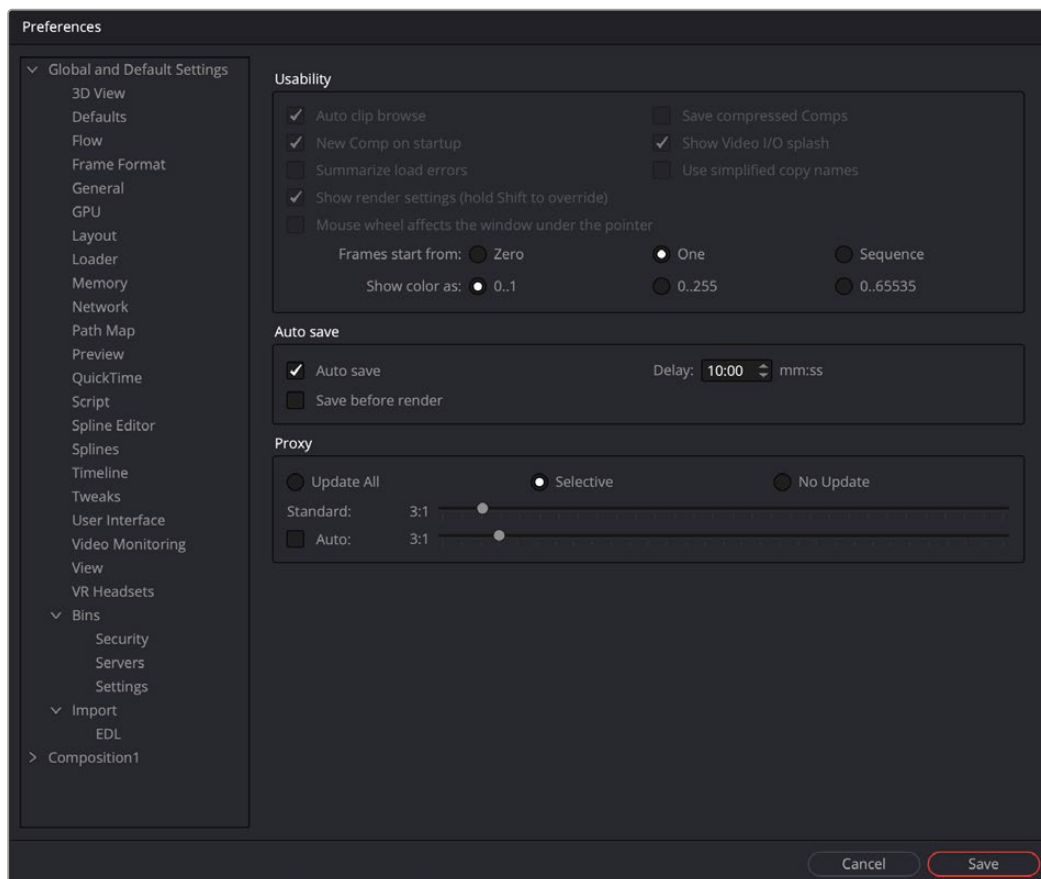
## Global and Composition Preferences

The Preferences window is divided into a category sidebar on the left and the settings panel on the right. In Fusion Studio, there are two levels of preferences: Global and Composition. The Fusion page in DaVinci Resolve uses just a single Global preference that affects every project, new and existing.

The Global preferences are used to set options specific to Fusion's overall behavior as well as defaults for each new composition. The Composition preferences in Fusion Studio can further modify the currently open composition without affecting the Global preferences or any other composition that is open but not displayed.

## Categories of Preferences

The first entry in the Preferences sidebar is assigned to the Global preferences. Clicking the Global and Default Settings disclosure arrow reveals the following sections.



Fusion Studio Global and Default settings

### 3D View

The 3D View preferences offer control over various parameters of the 3D Viewers, including grids, default ambient light setup, and Stereoscopic views.

### AVI (Windows Fusion Studio Only)

The AVI preferences configure the default compression options when creating AVI files from a Saver node. These settings can be overridden on a case-by-case basis using the Format tab in the Saver's Inspector.

### Defaults

The Defaults preferences are used to select default behavior for a variety of options, such as animation, global range, timecode display, and automatic tool merging.

## Flow

You use the Flow preferences to set many of the same options found in the Node Editor's contextual menu, like settings for Tile picture, the Navigator, and pipe style.

## Frame Format

The Frame Format preferences are used to create new frame formats as well as select the default image height and width when adding new creator tools like Background and Text+. You also set the frame rate for playback.

## General

The General preferences contain options for the general operation, such as auto save, and gamma settings for color controls.

## GPU (Fusion Studio Only)

The GPU preferences include options for selecting specific GPU acceleration methods based on your computer platform and hardware capabilities. It is also used for enabling caching, and debugging GPU devices and tools.

## Layout (Fusion Studio Only)

You can use the Layout preferences to save the exact layout of Fusion's windows.

## Loader (Fusion Studio Only)

Using the Loader preferences, you can set options for the default Loader's depth and aspect ratio as well as define the local and network LoaderCache settings.

## Memory (Fusion Studio Only)

Memory management for multi-frame and simultaneous branch rendering is configured in the Memory preferences.

## Network (Fusion Studio Only)

The Network rendering preferences are used to configure options such as selecting a render master, email notification, and whether the machine can be used as a render slave.

## Path Map

Path Map preferences are used to configure virtual file path names used by Loaders and Savers as well as the folders used by Fusion to locate comps, macros, scripts, tool settings, disk caches, and more.

## Preview (Fusion Studio Only)

The Preview preferences is where you configure the Preview creation and playback options.

## QuickTime (macOS Fusion Studio Only)

This section lets you preconfigure the QuickTime codec used for rendering.

## Script

The Script preferences include a field for passwords used to execute scripts externally, programs to use for editing scripts, and the default Python version to use.

## Spline Editor

The Spline Editor preferences allow you to set various spline options for Autosnap behavior, handles, markers, and more.

## Splines

Options for the handling and smoothing of animation splines, Tracker path defaults, onion-skinning, roto assist, and more are found in the Splines preference.

## Timeline

The Timeline preferences is where you create and edit Timeline/Spline filters and set default options for the Keyframes Editor.

## Tweaks

The Tweaks preferences handle miscellaneous settings for modifying the behavior when loading frames over the network and queue/network rendering.

## User Interface

These preferences set the appearance of the user interface window and how the Inspector is displayed.

## Video Monitoring (Fusion Studio Only)

The Video Monitoring preferences is where you can configure your Blackmagic video display hardware for monitoring on an HD, Ultra HD, or DCI 4K display.

## View

The View preferences are used to manage settings for viewers, including default control colors, Z-depth channel viewing ranges, default LUTs, padding for fit, zoom, and more.

## VR Headsets

The VR Headsets preferences allow configuration of any connected Virtual Reality headsets, including how stereo and 3D scenes are viewed.

## Bins (Fusion Studio Only)

There are three panels as part of the Bins preferences: a Security panel where you set users and passwords for serving the local bins; a Servers panel used to select which remote Bin servers are connected; and a Settings panel for stamp rendering.

## Import

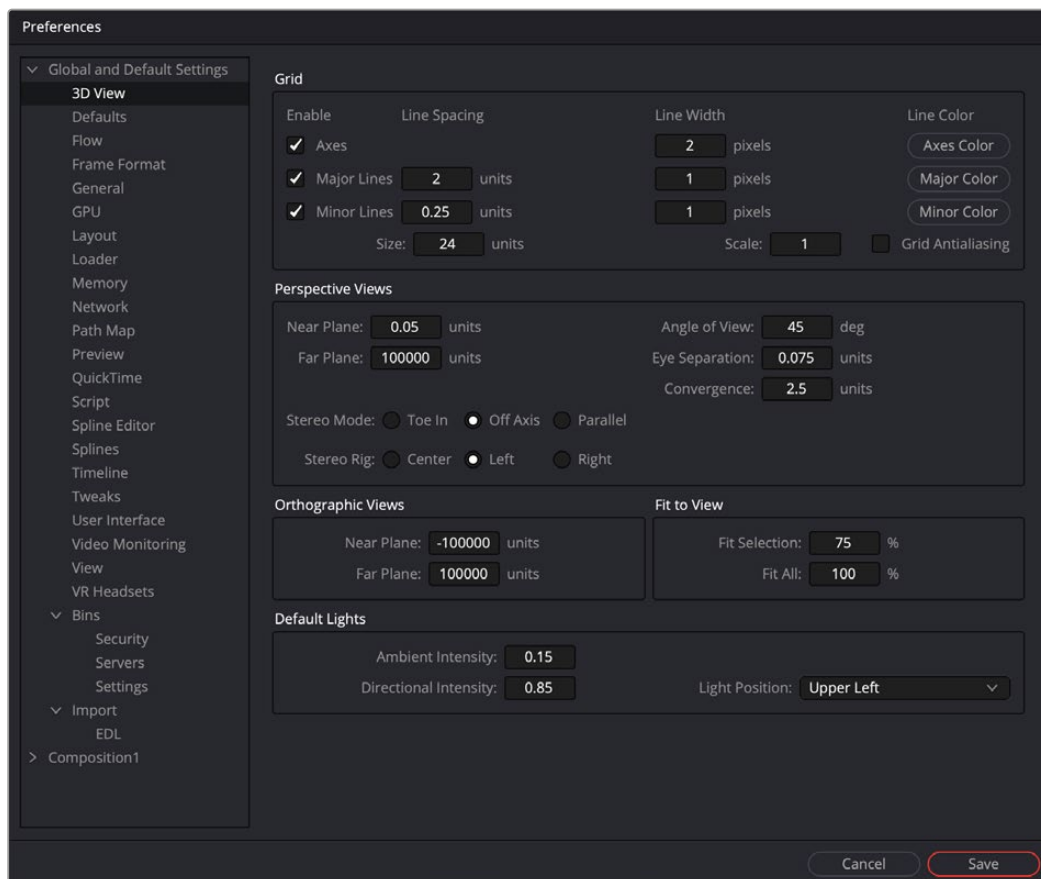
The Import settings contain options for EDL Import that affect how flows are built using the data from an EDL.

# Preferences In Depth

Within each category is a deep set of controls for configuring Fusion so that it better fits your working environment. The preferences contain both software and hardware options that affect all newly created comps. The following section explains every option located in the preferences categories.

## 3D View

The 3D View preferences contain settings for various defaults in the 3D Viewers, including grids, default ambient light setup, and Stereoscopic views.



The 3D View preferences

## Grid

The Grid section of the 3D View preferences configures how the grid in 3D Viewers are drawn.

- **Grid Antialiasing:** Some graphics hardware and drivers do not support antialiased grid lines, causing them to sort incorrectly in the 3D Viewer. Disabling this checkbox will disable antialiasing of the grid lines. To turn off the grid completely, right-click in a 3D Viewer and choose 3D Options > Grid.
- **Size:** Increasing the Size value will increase the number of grid lines drawn. The units used for the spacing between grid lines are not defined in Fusion. A “unit” is whatever you want it to be.
- **Scale:** Adjusting the overall scaling factor for the grid is useful, for example, if the area of the grid appears too small compared to the size of your geometry.

## Perspective Views

The Perspective Views section handles the appearance of the perspective view in both a normal and stereoscopic project.

- **Near Plane/Far Plane:** These values set the nearest and furthest point any object can get to or from the camera before it is clipped. The minimum setting is 0.05. Setting Near Plane too low and Far Plane too far results in loss of depth precision in the viewer.
- **Eye Separation/Convergence/Stereo Mode:** This group of settings defines the defaults when stereo is turned on in the 3D Viewer.

## Orthographic Views

Similar to the Perspective Views, the Orthographic Views (front, top, right, and left views) section sets the nearest and furthest point any object can get to or from the viewer before clipping occurs.

## Fit to View

The Fit to View section has two value fields that manage how much empty space is left around objects in the viewer when the F key is pressed.

- **Fit Selection:** Fit Selection determines the empty space when one or more objects are selected and the F key is pressed.
- **Fit All:** Fit All determines the empty space when you press F with no objects selected.

## Default Lights

These three settings control the default light setup in the 3D Viewer.

The default ambient light is used when lighting is turned on and you have not added a light to the scene. The directional light moves with the camera, so if the directional light is set to “upper left,” the light appears to come from the upper-left side of the image/camera.

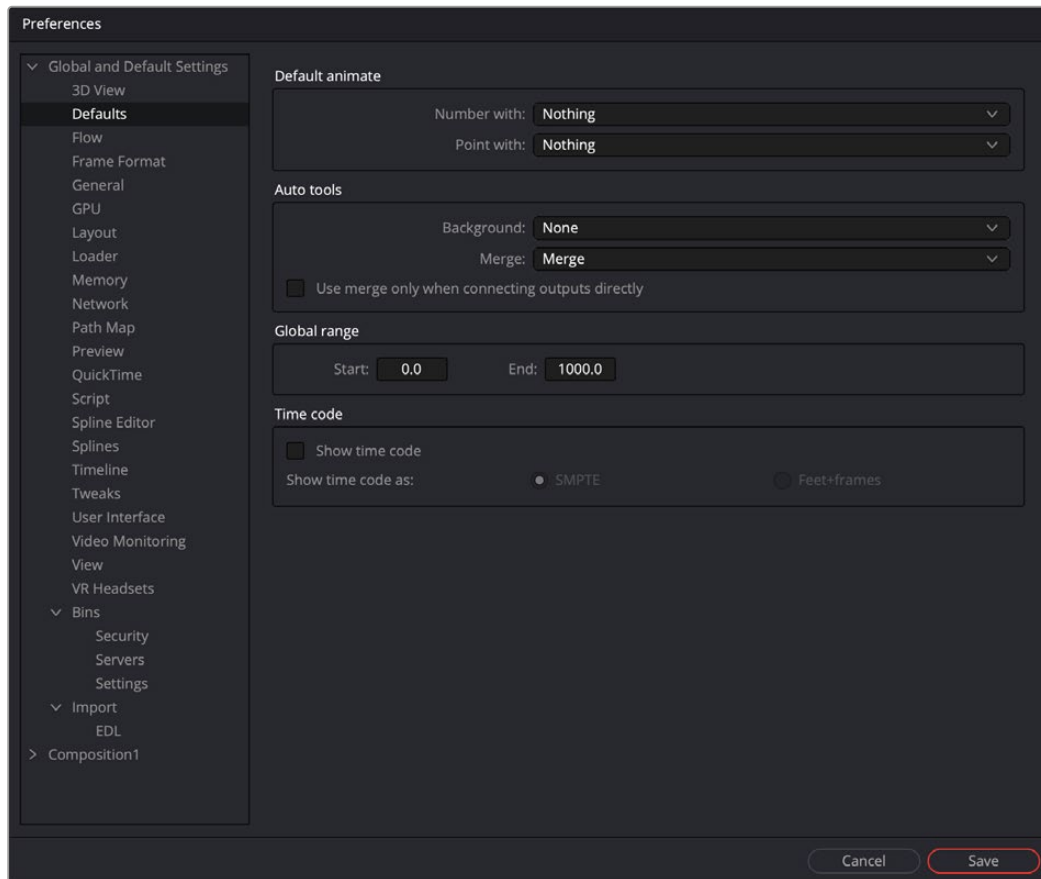
## AVI

The AVI preference is only available in Fusion Studio on Windows. It configures the default AVI codec settings when you select AVI as the rendering file format in the Saver node.

- **Compressor:** This drop-down menu displays the AVI codecs available from your computer. Fusion tests each codec when the application opens; therefore, some codecs may not be available if the tests indicate that they are unsuitable for use within Fusion.
- **Quality:** This slider determines the amount of compression to be used by the codec. Higher values produce clearer images but larger files. Not all codecs support the Quality setting.
- **Key Frame Every X Frames:** When checked, the codec creates keyframes at specified intervals. Keyframes are not compressed in conjunction with previous frames and are, therefore, quicker to seek within the resulting movie. Not all codecs support the keyframe setting.
- **Limit Data Rate To X KB/Second:** When checked, the data rates of the rendered file are limited to the amount specified. Not all codecs support this option. Enter the data rate used to limit the AVI in kilobytes (kB) per second, if applicable. This control does not affect the file unless the Limit Data Rate To option is selected.

## Defaults

The choices made here are used to determine Fusion's behavior when new tools are added to the Node Editor and when parameters are animated.



The Defaults preferences

### Default Animate

The Default Animate section is used to change the type of modifier attached to a parameter when the Animate option is selected from its contextual menu. The default option is Nothing, which uses a Bézier spline to animate numeric parameters and a path modifier for positional controls.

- **Number With and Point With:** Drop-down lists are used to select a different modifier for the new default. For example, change the default type used to animate position by setting the Point with the drop-down menu to XY Path.

Choices shown in this menu come from installed modifiers that are valid for that type of parameter. These include third-party plug-in modifiers, as well as native modifiers installed with Fusion.

### Auto Tools

The Auto Tools section determines which tools are added automatically for the most common operations of the Background tools and Merge operations.

- **Background:** When set to None, a standard Background tool is used; however, the drop-down menu allows you to choose from a variety of tools including 2D and 3D tools to customize the operation to your workflow.



- **Merge:** When set to None, nothing happens. When set to Merge, connecting the outputs of two tools or dragging multiple clips on the Node Editor uses a standard Merge. Other valid options for this are Anaglyph, Channel Booleans, and Dissolve.
- **Use Merge Only When Connecting Outputs Directly:** When this option is active, Merges are not automatically added when you drag multiple clips from the Finder or Windows Explorer onto the Flow area.

## Global Range

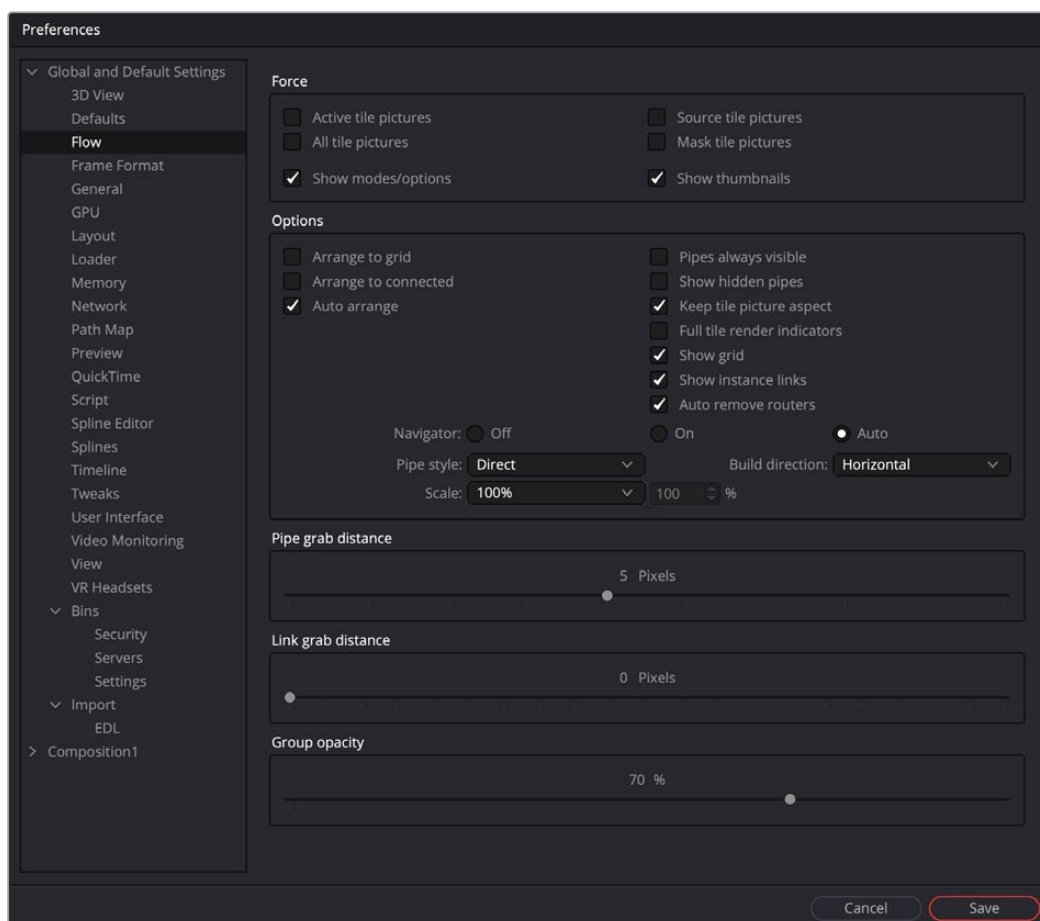
Using the Start and End fields, you can define the Global Start and End frames used when creating new compositions.

## Time Code

You use this option to determine whether new compositions will default to showing SMPTE Time Code or frames (Feet + Frames) to represent time.

## Flow

Many of the same options found in the Node Editor's contextual menu, like settings for Tile Picture, the Navigator, and Pipe Style, are found in this category.



The Flow preferences

## Force

The Force section can set the default to display pictures in certain tool tiles in the Node Editor rather than showing plane tiles. The Active checkbox sets pictures for the actively selected tool, the All checkbox enables pictures for all tiles, and the Source and Mask checkbox enables tile pictures for just Source and Mask tools.

When All is enabled, the picture shown will either be a thumbnail of the image rendered by the tool if the tool has rendered, or if the Show Thumbnails option is disabled, the tool's default icon is used. Concatenated transforms will also show a default icon.

- **Show Modes/Options:** Enabling this option will display icons in the tool tile depicting various states, like Disk Caching or Locked.
- **Show Thumbnails:** When this checkbox is selected, tool tiles set to show tile pictures will display the rendered output of the tool. When the checkbox is cleared, the default icon for the tool is used instead.

## Options

The Options section includes several settings that control or aid in the layout and alignment of tools in the Node Editor.

- **Arrange to Grid:** This enables a new node tree's Snap to Grid option to force the tool layout to align with the grid marks in the flow.
- **Arrange to Connected:** Tools snap to the vertical or horizontal positions of other tools they are connected to.
- **Auto Arrange:** This option enables the Node Editor to shift the position of tools as needed to make space when inserting new tools or auto-merging layers.
- **Show Grid:** This enables or disables the display of the Node Editor's background grid.
- **Auto Remove Routers:** Pipe Routers or "elbow nodes" in the Node Editor are considered "orphaned" if the tools connected to either the input or output are deleted. When this option is enabled, Orphaned Routers are automatically deleted.
- **Pipes Always Visible:** When enabled, the connection lines between tools are drawn over the top of the tool tiles.
- **Keep Tile Picture Aspect:** Enabling this option forces tool tile thumbnail pictures to preserve the aspect of the original image in the thumbnail.
- **Full Tile Render Indicators:** Enabling this checkbox causes the entire tile to change color when it is processing. This can make it easier to identify which tools are processing in a large composition. The coloring itself will form a progress bar to alert you to how close slower tools are to finishing their process.
- **Show Instance Links:** This option is used to select whether Instance tools will show links, displayed as green lines, between Instance tools.
- **Navigator:** The Navigator is a small square overview of the entire composition. It is used to quickly navigate to different parts of a node tree while you are zoomed in. The checkboxes in this section determine when the Navigator is displayed, if at all.
  - On:** The Navigator will always be visible.
  - Off:** The Navigator will always be hidden.
  - Auto:** The Navigator will only be visible when the Node Editor's contents exceed the currently visible Work area.
- **Pipe Style:** This drop-down menu selects which method is used to draw connections between tools. The Direct method uses a straight line between tools, and Orthogonal uses horizontal and vertical lines.

- **Build Direction:** When auto-building or laying out a node tree, Build Direction controls whether tools are organized horizontally or vertically.
- **Scale:** The Scale menu allows you to select the default zoom level of the Node Editor when a new composition is created.

## Pipe Grab Distance

The Pipe Grab Distance slider allows you to choose how close the pointer must be (in pixels) to a connection line in the node tree when selecting them.

## Link Grab Distance

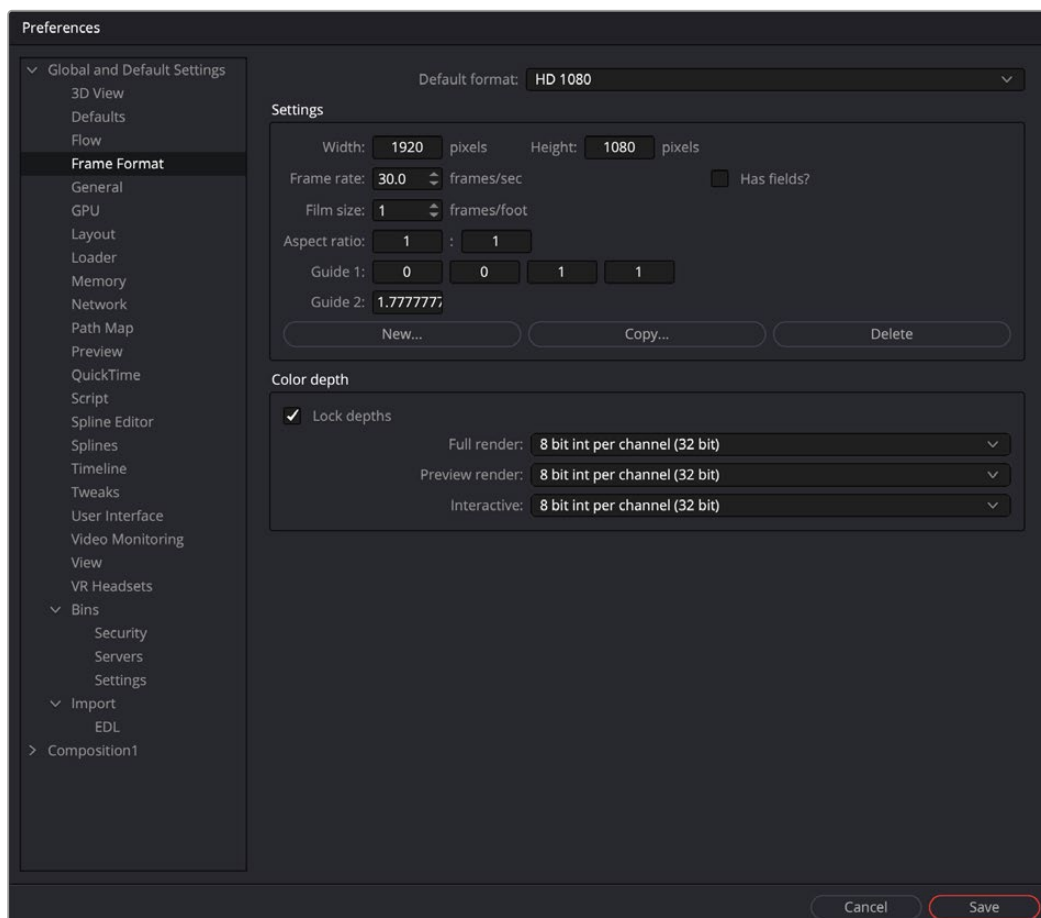
The Link Grab Distance slider allows you to choose how close the pointer must be (in pixels) to a knot on a node before a connection is made or removed. 0 pixels means you must be directly on the knot, while a maximum setting of 20 means you can be 20 pixels from a knot and still connect to it.

## Group Opacity

This slider controls the opacity of an expanded group's background in the Node Editor.

## Frame Format

Frame Format preferences allow you to select the resolution and frame rate for the nodes that generate images like Background, fast noise, and Text+. It also sets the color bit depth for final renders, previews, and interactive updates in the viewer. The color bit depth settings only apply to Fusion Studio. Rendering in DaVinci Resolve always use 32-bit float.



The Frame Format preferences

## Default Format

This drop-down menu is used to select the default resolution for Generator tools from a list of presets. This is only a default setting; these settings can be overridden using the Resolution settings in a node's Inspector.

Use the Edit boxes to change any of the default settings. When creating a new setting, press the New button and enter a name for the setting in the dialog box that appears and enter the parameters.

## Settings

The Settings section defines the format that is selected in the Default Format menu. You can modify an existing format or create a new one.

- **Width/Height:** When creating a new format for the menu or modifying an existing menu item, you specify the Width or Height in pixels of the format using these fields.
- **Frame Rate:** Enter or view the frames per second played by the format. This sets the default Frame Rate for previews and final renders from the Saver tool. It also sets the playback for the comp itself, as well as the frame to time code conversion for tools with temporal inputs.
- **Has Fields:** When this checkbox is enabled, any Creator or Loader tool added to the Node Editor will be in Fields process mode.
- **Film Size:** This field is used to define how many frames are found in one foot of film. The value is used to calculate the display of time code in Feet + Frames mode.
- **Aspect Ratio:** These two fields set the pixel aspect ratio of the chosen frame format.
- **Guide 1:** The four fields for Guide 1 define the left, top, right, and bottom guide positions for the custom guides in the viewer. To change the position of a guide, enter a value from 0 to 1. The bottom-left corner is always 0/0, the top-right corner is always 1/1. If the entered value's aspect does not conform to the frame format as defined by the Width and Height parameters, an additional guide is displayed onscreen. The dotted line represents the image aspect centered about Guide 1's Center values.
- **Guide 2:** This setting determines the image aspect ratio in respect to the entire frame format width and height. Values higher than 1 cause the height to decrease relative to the width. Values smaller than 1 cause height to increase relative to width.
- **New:** You use the New button to create a new default setting in the drop-down menu. Once you click the button, you can name the setting in the dialog box that appears.
- **Copy:** The Copy button copies the current setting to create a new one for customization.
- **Delete:** The Delete button will remove the current setting from the default drop-down list.

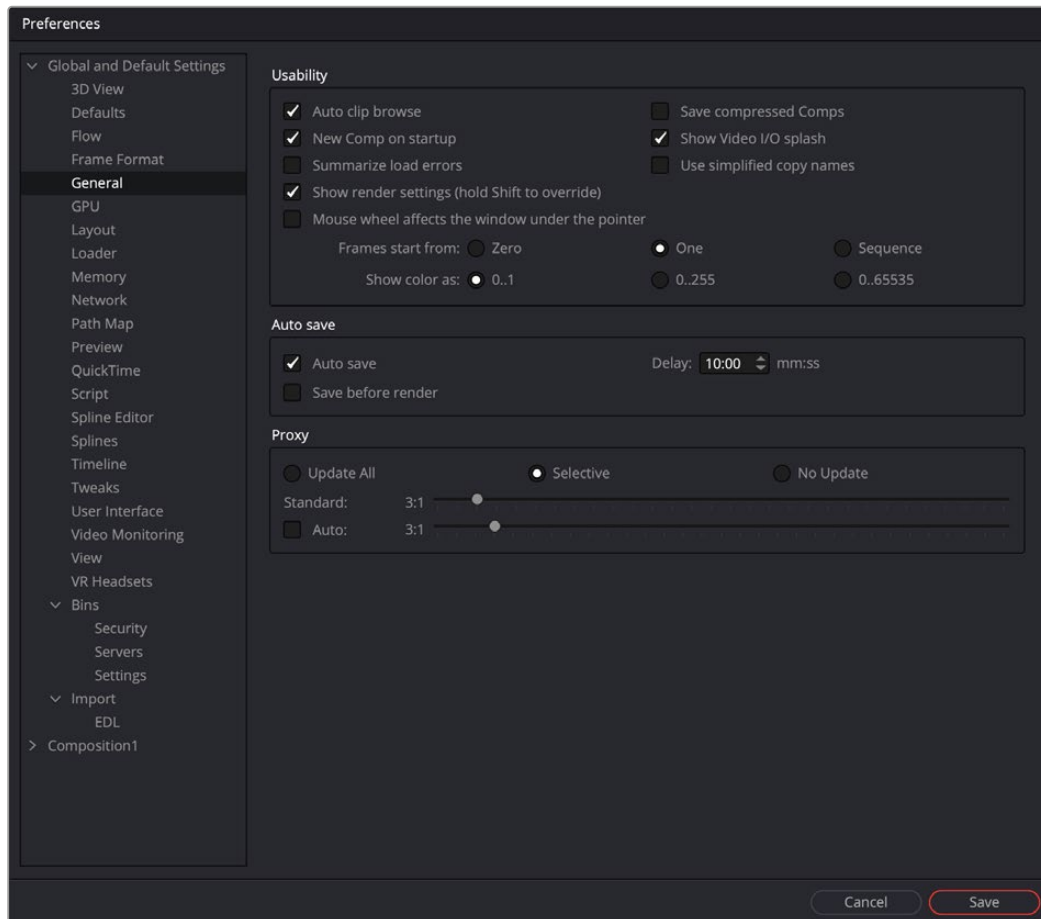
## Color Depth

The three menus in the Color Depth section are used to select the color mode for processing preview renders, interactive renders, and full (final) renders. Processing images at 8-bit is the lowest color depth and is rarely sufficient for final work these days but is acceptable for fast previews. 16-bit color has much higher color fidelity but uses more system resources. 16-bit and 32-bit float per channel uses even more system resources and is best for digital film and HDR rendered images.

Generally, these options are ignored by the composition unless a Loader or Creator tool's Color Depth control is set to Default.

## General

The sections contained in the General preferences affect the behavior of the Inspector as well as some other user interface elements.



The General preferences

## Usability

Usability has a number of project, Node Editor, and user interface settings that can make the application easier to work with, depending on your workflow.

- **Auto Clip Browse:** When this checkbox is enabled, the File Browser is automatically displayed when a new Loader or Saver is added to the Node Editor.
- **New Comp on Startup:** When checked, a new, empty project is created each time Fusion Studio is launched. This has no effect in DaVinci Resolve's Fusion page.
- **Summarize Load Errors:** When loading node trees or “comps” that contain unknown tools (e.g., comps that have been created on other computers with plugins not installed on the current machine), the missing tools are summarized in the console rather than a dialog being presented for every missing tool.
- **Save Compressed Comps:** This option enables the saving of compressed node trees, rather than ASCII based text files. Compressed node trees take up less space on disk, although they may take a moment longer to load. Node trees containing complex spline animation and many paint strokes can grow into tens of megabytes when this option is disabled. However, compressed comps cannot be edited with a text editor unless saved again as uncompressed.

- **Show Video I/O Splash:** This toggles whether the Splash image will be displayed over the video display hardware. This only applies to Fusion Studio.
- **Use Simplified Copy Names:** This option reduces the occurrence of underscores in tool names when copying.
- **Show Render Settings:** When this checkbox is selected, the Fusion Render Settings dialog will be displayed every time a render is started in Fusion Studio. Holding Shift while starting a render will prevent the display of the dialog for that session, using whatever settings were applied during the last render. Disabling this option reverses this behavior.
- **Mouse Wheel Affects the Window Under the Pointer:** Normally the mouse wheel or trackpad swiping works in the currently active window. With this option enabled, it will work in the window underneath the cursor, so you don't have to click into a window first to make it active.
- **Frames Start From:** This designates the starting frame number for clip times in the Loader and its Clip list.
- **Show Color As:** This setting determines the numeric scale used to represent colors. The available options are Normalized (0 to 1), 8-bit (0 to 255), and 16-bit (0 to 65,535). This does not affect the actual processing or quality of the image, but it can make the mental math sometimes used to figure out adjustments a bit easier.

## Auto Save

The Auto Save settings only apply to Fusion Studio. To set auto backups for the Fusion page in DaVinci Resolve, use the DaVinci Resolve Project Load and Save Preferences.

When Auto Save is enabled in Fusion Studio, comps are automatically saved to a backup file at regular intervals defined by the Delay setting. If a backup file is found when attempting to open the comp, you are presented with the choice of loading either the backup or the original.

If the backup comp is opened from the location set in the Path Map preference, saving the backup will overwrite the original file. If the backup file is closed without saving, it is deleted without affecting the original file.

- **Save Before Render:** When enabled, the comp is automatically saved before a preview or final render is started.
- **Delay:** This preference is used to set the interval between Auto Saves. The interval is set using mm:ss notation, so entering 10 causes an Auto Save to occur every 10 seconds, whereas entering 10:00 causes an Auto Save every 10 minutes.

## Proxy

- **Update All, Selective, No Update:** The Update mode button is located above the toolbar. You can use this preference to determine the default mode for all new comps. Selective is the usual default. It renders only the tools needed to display the images in the Display view. All will render all tools in the composition, whereas None prevents all rendering.
- **Standard and Auto:** These sliders designate the default ratio used to create proxies when the Proxy and Auto Proxy modes are turned on. These settings do not affect the final render quality.

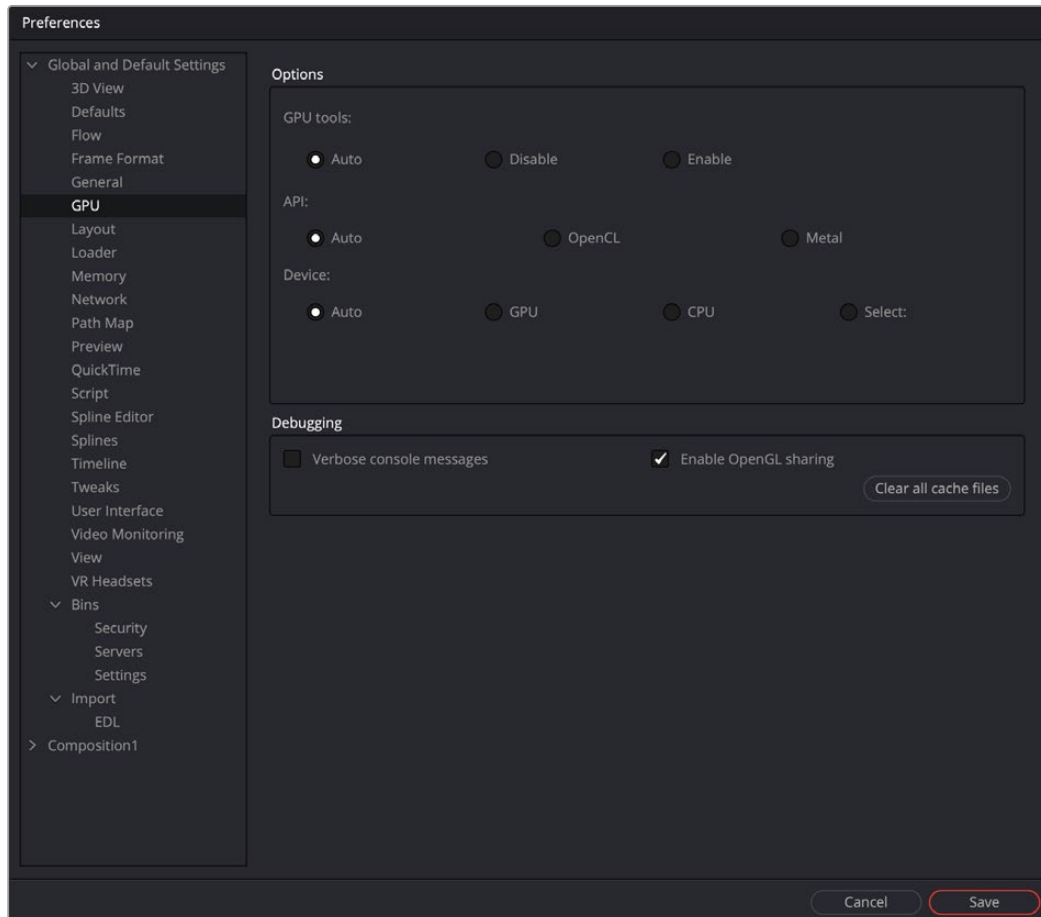
Even though the images are being processed smaller than their original size, the image viewing scales in the viewers still refer to original resolutions. Additionally, image processing performed in Proxy Scale mode may differ slightly from full-resolution rendering.

The Proxy and Auto Proxy size ratios may be changed from within the interface itself by right-clicking on the Prx and APrx buttons above the toolbar and selecting the desired value from the contextual menu.

## GPU

The GPU preference is only available in Fusion Studio. In DaVinci Resolve, you can configure the GPU processing in Resolve's Memory and GPU preferences.

In Fusion Studio, the GPU preference is used to specify the GPU acceleration method used for processing, based on your computer platform and hardware capabilities. It is also used for enabling caching and debugging GPU devices and tools.



The GPU preferences

### Options

The GPU options include radio buttons to select whether the GPU is used when processing and, if so, which computer framework is used for communicating with the GPU.

- **GPU Tools:** This preference has three settings: Auto, Disable, and Enable. When set to Disable, no GPU acceleration is used for tools or third-party plugins. Fuses may still require GPU acceleration. If Enable is selected, GPU acceleration is available for tools and plugins, if appropriate drivers are installed.
- **API:** The API setting selects the GPU processing method to use.
- **Device:** The Device setting determines which GPU hardware to use in the case of multiple GPUs. The Auto setting gives priority to GPU processing; however, if it is unavailable, Fusion uses the platform default. Currently, both the AMD and CPU options require either the AMD Catalyst 10.10 Accelerated Parallel Processing (APP) technology Edition driver or the ATI Stream SDK 2.1 or later to be installed. The Select setting allows you to choose the device explicitly.

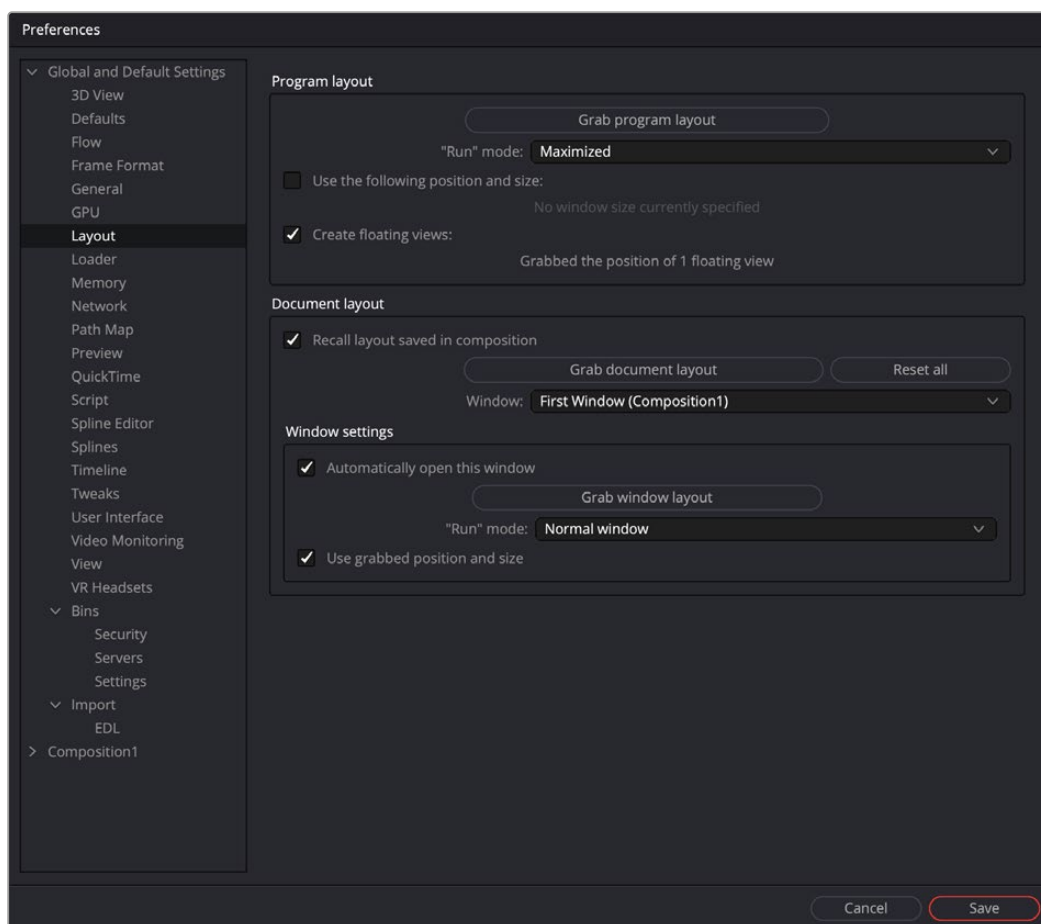
## Debugging

The more advanced preferences located in this section are designed for diagnostics and analyzing GPU operations.

- **Verbose Console Messages:** Enabling this option causes information to be shown in the Console. For example, Startup Logs, Compiler Warnings, and Messages.
- **OpenGL Sharing:** Enabling this option shares system RAM with onboard GPU RAM to create a larger, but slower, OpenGL memory pool.
- **Clear Cache Files:** This option will clear already compiled GPU code and then recompile the kernels.

## Layout

The Layout preferences are only available in Fusion Studio. To save a Layout in DaVinci Resolve's Fusion page, use the Workspace > Layout Presets menu. The Layout options are used to control the layout, size, and position of various windows in Fusion's interface at startup or when a comp is created.



The Layout preferences

There are a lot of options, but in practice, you simply organize the interface the way you prefer it on startup and when a new composition is created, then open this Preferences panel and click on the three buttons to grab the Program Layout, the Document Layout and the Window Settings.



## Program Layout

The Program Layout is used to save the overall Fusion interface window and any open floating windows. Each new composition you open within the larger overall Fusion interface window will adhere to these preferences.

- **Grab Program Layout:** Pressing this button stores the application's overall current position and size.
- **Run Mode:** This menu is used to select the application's default mode at startup. You choose between a Maximized application window, a Minimized application, or a Normal application display.
- **Use the Following Position and Size:** When checked, the values stored when Grab Program Layout was selected will be used when starting Fusion Studio.
- **Create Floating Views:** When checked, the position and size of the floating viewers will be saved when the Grab Program Layout button is used.

## Document Layout

The Document Layout is used to save the layout of panels and windows for the current Fusion comp.

- **Recall Layout Saved In Composition:** When checked, all Document Layout settings in the controls below will be recalled when a saved composition is loaded.
- **Grab Document Layout:** Pressing this button stores the entire interface setup, including all the internal positions and sizes of panels and work areas.
- **Window:** When multiple windows on the same composition are used, this menu is used to select the window to which the Window Settings will apply.

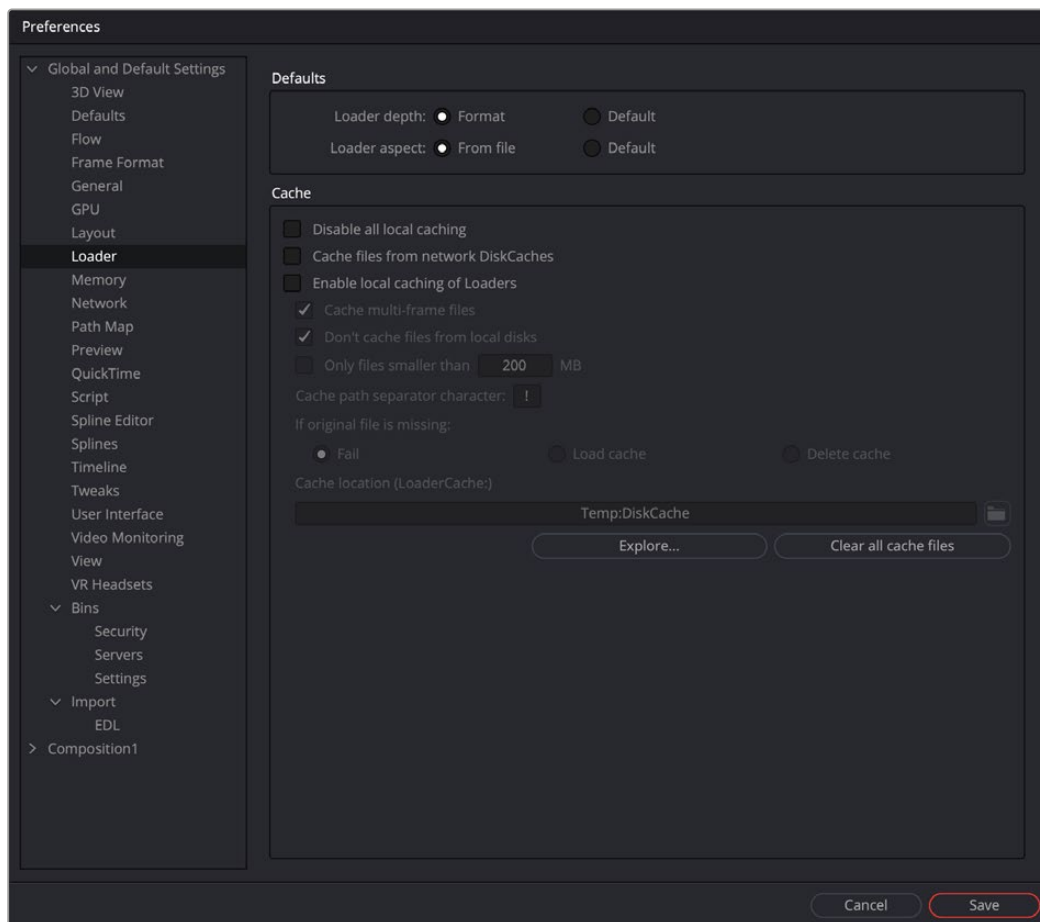
## Window Settings

Rather than saving entire comp layouts, you can save position and size for individual floating windows and panels within a comp using the Window Settings.

- **Automatically Open This Window:** When checked, the selected window will automatically be opened for new flows.
- **Grab Window Layout:** Pressing this button stores the size and position of the selected window.
- **Run Mode:** Select the default run mode for the selected window. You can choose between a Maximized window, a Minimized window, or a Normal window display.
- **Use Grabbed Position and Size:** When checked, the selected window will be created using the stored position and size.

## Loader

The Loader preferences are only available in Fusion Studio. Using the Loader preferences, you can set options for the default Loader's color depth and aspect ratio as well as define the local and network cache settings.



The Loader preferences

## Defaults

The Defaults section includes two settings to determine how color depth and aspect ratio are handled for Loaders.

- **Loader Depth:** The Loader Depth defines how color bit depth is handled when adding a Loader. Choosing Format means that the correct bit depth is automatically selected, depending on the file format and the information in the file's header. Choosing Default sets the bit depth to the value specified in the Frame Format preferences.

## Cache

The Cache preferences allow you to control how disk caching operates in Fusion. You can set how and where the cache is generated, when the cache is removed, how the cache reacts when source files are not available, as well as many other cache related options. This is not to be confused with RAM cache, which is controlled in the Memory preferences.

- **Disable All Local Caching:** This setting disables local caching.
- **Cache Files from Network DiskCaches:** If a tool has disk caching enabled, and the disk cache files are stored remotely on the network, then enabling this option will use a local copy of those cache files, similarly to the local cache on a networked Loader.
- **Enable Local Caching of Loaders:** Files will be copied into the LoaderCache path set below or in the Path Maps preferences.

- **Cache Multi-Frame Files:** Files like AVI or QuickTime will be copied into the LoaderCache path. This may take some time if the file is large.
- **Don't Cache Files from Local Disks:** Files that do not sit on a network drive will not be copied into the LoaderCache path. You can disable this option if you have, for example, a fast SSD cache drive and want to use it for local files as well to speed up file access while working interactively.
- **Only Files Smaller Than xxx MB.:** Files larger than the value set here will not be copied into the LoaderCache path.
- **Cache Path Separator Character:** When Enable Local Caching of Loaders is enabled, you can use this setting to rebuild the path of the original files in LoaderCache.  
For instance, given the default “!” character, the original path X:\Project\MyShots\Shot0815\ will be translated into X!Project!MyShots!Shot0815! in the LoaderCache path. Other separator characters may be used, including the “\” character, which will use subdirectories in LoaderCache: X\Project\MyShots\Shot0815\.
- **If Original File Is Missing:** This setting provides three options to determine the caching behavior when the original files can't be found. The Fail option behaves exactly as the Default Loader in Fusion. The Loader will not process, which may cause the render to halt. The Load Cache option loads the cache even though no original file is present. The Delete Cache option clears missing files from the cache.
- **Cache Location:** For convenience, this is a copy of the LoaderCache path set in the Path Maps preferences.
- **Explore:** This button opens the LoaderCache path in the macOS X Finder window or a Windows Explorer window.
- **Clear All Cache Files:** This button deletes all cached files present in the LoaderCache path.

## Memory

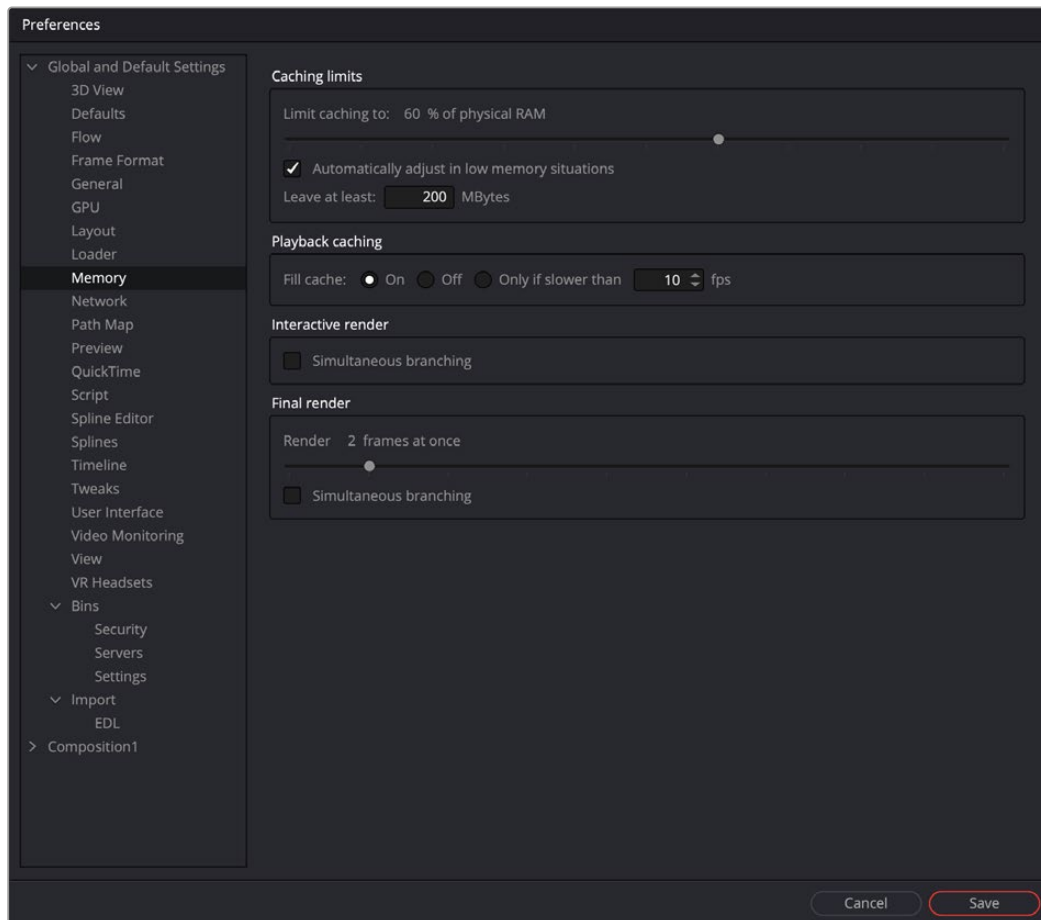
The Memory preferences are only available in Fusion Studio. To control Fusion's memory when using the Fusion page in DaVinci Resolve, open DaVinci Resolve's Memory and GPU preferences.

Occasionally, it will be necessary to adjust the Memory preferences in order to make the best use of available memory on the computer. For example, some people prefer a higher cache memory for faster interactive work, but for final renders the cache memory is often reduced, so there's more memory available for simultaneous processing of tools or multiple frames being rendered at once.

### Caching Limits

The Caching Limits include options for Fusion's RAM cache operation. Here, you can determine how much RAM is allocated to the RAM cache for playing back comps in the viewer.

- **Limit Caching To:** This slider is used to set the percentage of available memory used for the interactive tool cache. Available memory refers to the amount of memory installed in the computer. When the interactive cache reaches the limit defined in this setting, it starts to remove lower priority frames in the cache to clear space for new frames.
- **Automatically Adjust In Low Memory Situations:** This checkbox will set the caching to adjust when memory is low. The console will display any cache purges.
- **Leave At Least X MBytes:** This setting is used to set the hard limit for memory usage. No matter what the setting of the Cache Limit, this setting determines the amount of physical memory available for use by other applications. Normally, this value should not be smaller than 25 MBytes.



The Memory preferences

## Interactive Render

The Interactive Render option allows you to optimize Fusion's processing based on the amount of RAM you have installed in your system.

- **Simultaneous Branching:** When checked, more than one tool will be processed at the same time. Disable this checkbox if you are running out of memory frequently.

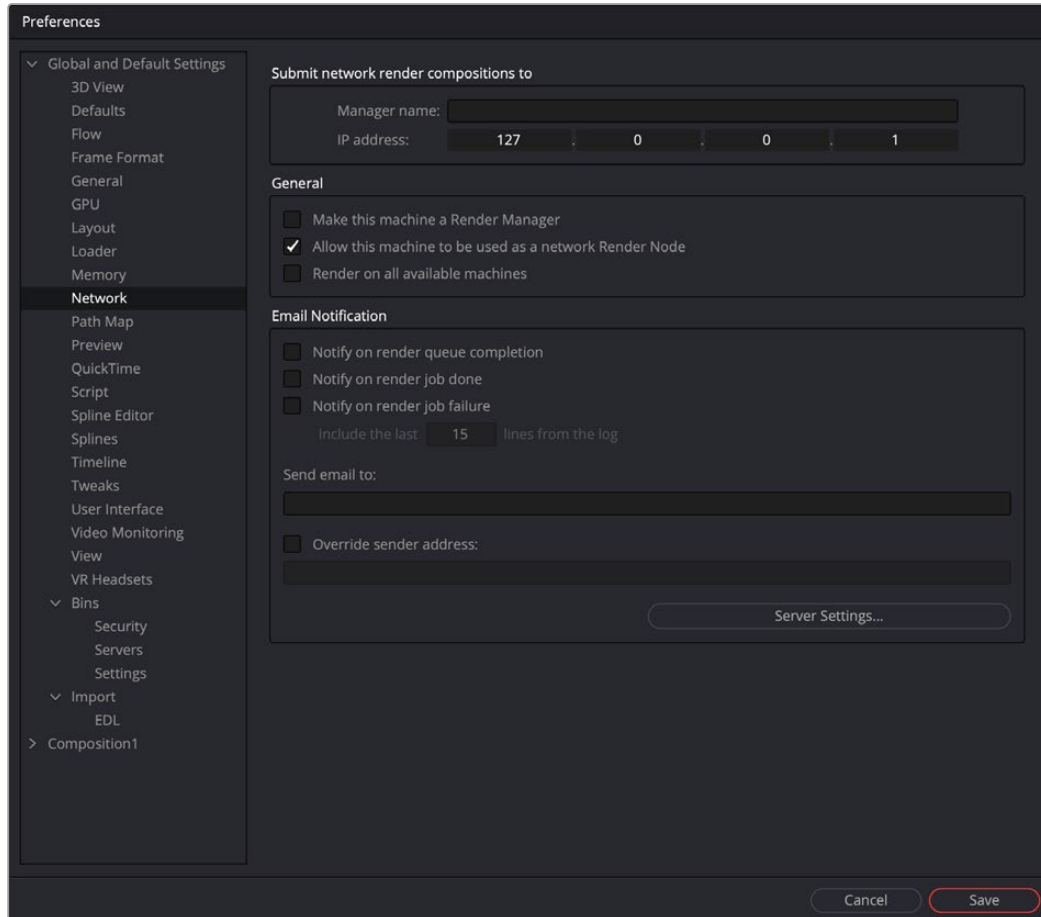
## Final Render

These settings apply to memory usage during a rendering session, either preview or final, with no effect during an interactive session.

- **Render Slider:** This slider adjusts the number of frames that are rendered at the same time.
- **Simultaneous Branching:** When checked, more than one branch of a node tree will be rendered at the same time. If you are running low on memory, turn this off to increase rendering performance.

## Network

The Network preferences are only available in Fusion Studio. These preferences are used to set up and control network rendering in Fusion Studio. The majority of settings are found in the Render Manager dialog.



The Network preferences

### Submit Network Render Compositions

In these fields, you enter the Master Name and IP address of the computer that will manage all network renders sent from this machine. If a standalone render master is in use on the network, these fields may be pre-filled and may not be editable. This is done to prevent multiple unauthorized render masters from being created by each person in a facility.

To re-enable editing of the master name and IP, create the environment variable `FUSION_NO_MANAGER` and set the value to `True`. Check your operating system user guide for how to create environment variables.

## General

The General preferences are designed with the most used options at the top in the General section. These options determine in what capacity the system is used during network rendering.

- **Make This Machine a Render Master:** When enabled, Fusion will accept network render compositions from other computers and manage the render. It does not necessarily mean that this computer will be directly involved in the render, but it will submit the job to the render nodes listed in the Render Manager dialog.
- **Allows This Machine to Be Used as a Network Slave:** When enabled, this computer can be used as a Render node and will accept compositions for network rendering. Deselect it to prevent other people from submitting compositions to render on this computer.
- **Render on All Available Machines:** Enable this checkbox to ignore groups and priorities configured in the Render Manager. Compositions submitted from this computer for network rendering will always be assigned to every available slave.

## Email Notification

You can use the Email Notification section to set up who gets notified with status updates regarding the render jobs and the network.

- **Notify Options:** These checkboxes cause emails to be sent when certain render events take place. The available events are Queue Completion, Job Done, and Job Failure.
- **Send Email to:** Enter the address or addresses to which notifications should be sent. You separate multiple addresses with a semicolon.
- **Override Sender Address:** Enter an email address that will be used as the sender address. If this option is not selected, no sender address is used, which may cause some spam filters to prevent the message from being delivered to the recipient.

## Server Settings

This section covers Clustering and Network Rendering.

For more information on these settings and clustering, see *Chapter 66, "Rendering Using Saver Nodes,"* in the DaVinci Resolve Reference Manual or *Chapter 4* in the Fusion Reference Manual.

## Path Maps

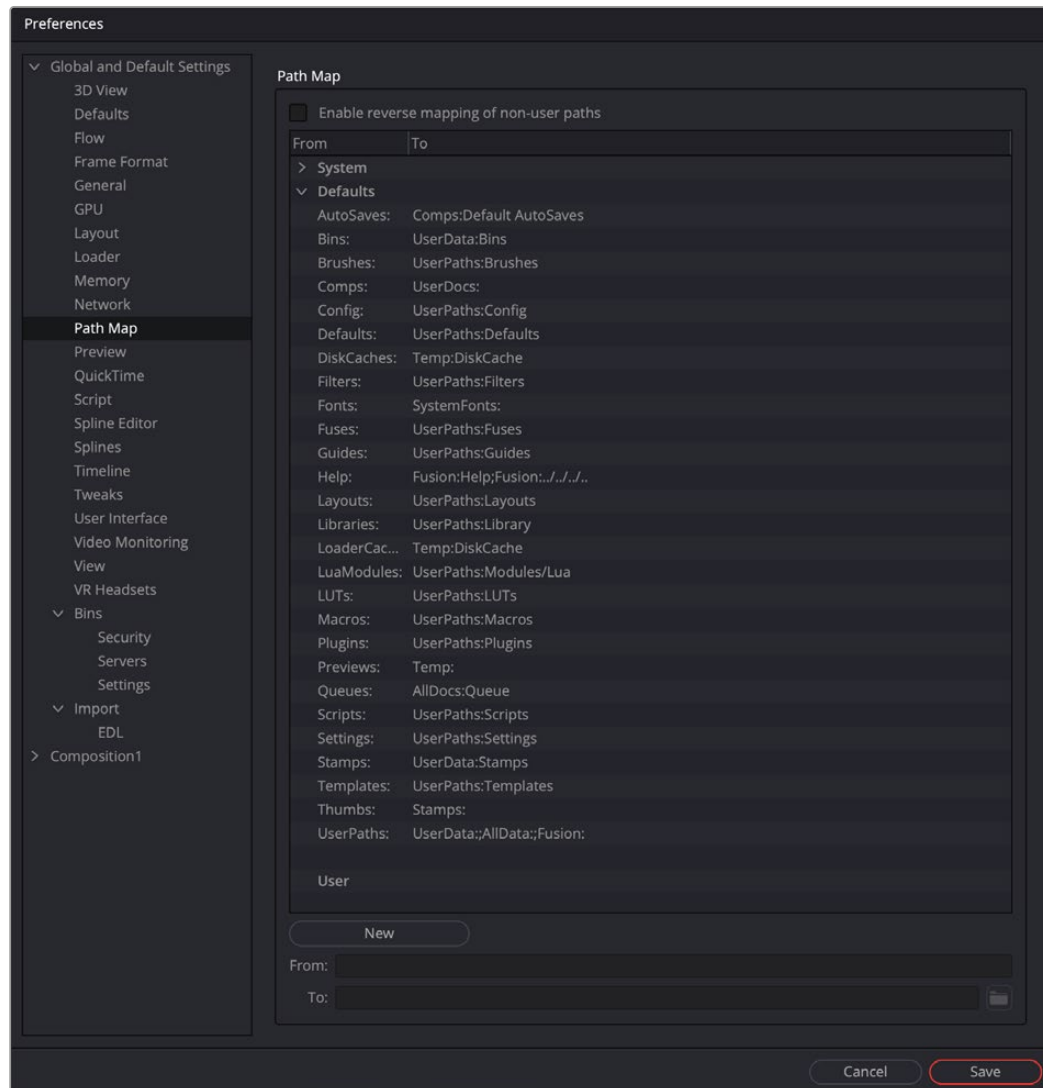
Path Maps are virtual paths used to replace segments of file paths with variables. For example, define the path 'movie\_x' as actually being in X\Shows\Movie\_X. Using this example, Fusion would understand the path 'movie\_x\scene\_5\scan.000.cin' as actually being X\Shows\Movie\_X\scene\_5\scan.000.cin.

For Fusion Studio, there are two main advantages to virtual path maps instead of actual file paths. One is that you can easily change the path to media connected to Loaders (for example, when moving a comp from one drive to another), without needing to make any changes in the composition. The other advantage is when network rendering, you can bypass the different OS filename conventions.

- **Enable Reverse Mapping of Paths Preferences:** This checkbox is at the top of the Path Map settings. When enabled, Fusion uses the built-in path maps for entries in the path's settings when applying mapping to existing filenames. The main benefit is for Fusion Studio. Enabling this checkbox causes Loaders to automatically use paths relative to the location of the saved composition when they are added to the Node Editor.

For more information on using relative paths for Loaders, see *Chapter 105, “I/O Nodes,”* in the DaVinci Resolve Reference Manual or Chapter 45 in the Fusion Reference Manual.

As with other preferences in Fusion Studio, paths maps are available in both Global and Composition preferences. Global preferences are applied to all new compositions, while Composition path maps are only saved with the current composition. Composition path maps will override Global path maps with the same name.



The Path Map preferences

The Global paths maps are divided into three sections:

- **System Path Maps:** The operating system determines system path maps, and they define Fusion’s global locations. You can override specific System path maps using the Defaults or current Composition Path Map settings. If you change your mind at a later time, you are always able to return to Fusion’s “factory” defaults using the System path maps. There are several top-level path maps established in the System Path Map settings.

**AllData:** The folder where Fusion saves all shared application data.

**AllDocs:** The folder where Fusion saves the public/shared document folder.

**AllLUTs:** The nested LUTs path in the Defaults section, where Fusion saves LUTs.

**Fusion:** The folder where Fusion Studio app is installed. For example, if you open Fusion from C:\Program Files\Fusion, then the path Fusion:\Help refers to C:\Program Files\Fusion\Help. If you instead used a copy of Fusion found in \\post-server\fusion\16, then Fusion:\Help would expand to \\post-server\fusion\16\Help.

**FusionLibs:** The Fusion libraries used for the application.

**Profile:** The folder where default Fusion preferences file is saved.

**Profiles:** The folder where Fusion individual user preferences are saved.

**Programs:** The location of Fusion Studio or DaVinci Resolve.

**SystemFonts:** The folder where the OS saves fonts that appear for Text+ and Text 3D nodes.

**Temp:** The system's temporary folder.

**UserData:** The folder where Fusion saves all user-specific miscellaneous roaming data. The individual elements included in the roaming data are listed in the Default Path Maps section. For Fusion Studio on Windows, this is "C:\Users\username\AppData\Roaming\". On Linux, this will be "\$HOME/Blackmagic/Fusion". On macOS, this is "Users/UserName/Library/Application Support/Blackmagic Design/Fusion".

**UserDocs:** The folder where Fusion saves the user's document folders.

- **Default Path Maps:** The Defaults are user-editable path maps. They can reference the System paths, as part of their paths. For instance, the Temp folder is defined in the System path and used by the Default DiskCache path map to refine the nested location (Temp:DiskCache). Default path maps can also redirect paths without using the Global System path maps. After you change a Default, the updated setting can be selected in the Preferences window, and a Reset button at the bottom of the Preferences window will return the modified setting to the System default.

**AutoSaves:** This setting determines the Fusion Comp AutoSave document's location, set in the Fusion General preferences.

**Bins:** Sets the location of Fusion Studio bins. Since the bins use pointers to the content, the content is not saved with the bin. Only the metadata and pointers are saved in the bins.

**Brushes:** Points Fusion to the folder that contains custom paintbrushes.

**Comps:** The folder where Fusion Studio compositions are saved. On macOS or Windows, the default location is in Users/YourUserName/Documents/Blackmagic Design/Fusion.

**Config:** Stores Configuration files used by Fusion Studio during its operation.

**Defaults:** Identifies the location of node default settings so they can be restored if overwritten.

**DiskCache:** Sets the location for files written to disk when using the Cache to Disk feature. This location can be overridden in the Cache to Disk window.

**Edit templates:** The location where Fusion macros are saved in order to appear as templates in the DaVinci Resolve Effects Library.

**Filters:** Points to a folder containing Convolution filters like sharpen, which can be used for the Custom Filter node.

**Fonts:** The default path map for Fonts points to the operating system fonts folders. Changing this will change the fonts that are available in the Text+ or Text 3D nodes as well as any Fusion Title Template. In DaVinci Resolve. This path map does not affect the five additional Edit page titles (L Lower 3rd, R Lower 3rd, M Lower 3rd, Scroll, and Text.)

**Fuses:** Points to a folder containing Fusion Fuses plugins.

**FusionTemplates:** Location where Fusion macros are saved in order to appear as templates in Fusion's Effects Library.



**Guides:** Location where custom viewer guide overlays are stored.

**Help:** Identifies where Fusion Studio PDF files are located.

**Layouts:** Location where Fusion Studio custom window layouts are saved.

**Libraries:** Points to a support folder where custom Effects Library items can be stored.

**LoaderCache:** The Fusion Studio Loader preferences allow the Loader to cache when reading from a slow network. This path map point to the local drive location for that cache.

**LuaModules:** Location for Lua Scripting modules.

**LUTs:** Points to a folder containing Look Up Tables (LUTs).

**Macros:** Points to the location for user created macros. The macros saved to this location appear in the macros category of the Effects Library and in the right-click Edit Macro contextual menu.

**Plugins:** This refers to user specific OpenFX plugins that you do not want loaded for all users.

**Previews:** Path map used for the older style, file sequence flipbook previews.

**Queues:** Location of the Render manager list.

**Scripts:** Location of Lua and Python scripts. This path can be further refined into specific scripts for tools (nodes), comps, and other specific script types.

**Settings:** Location where custom Node settings are saved.

**Stamps:** Location for preview movies generated in a Fusion Studio bin. This is an outdated path map since bins now include the Studio Player.

**Templates:** Location of the Templates folder. Saving Macros to the Template folder will cause them to appear in the Effects Library in a Templates category. In Fusion Studio, the Templates category does not appear until a Macros is saved into the folder.

**Thumbs:** Location for clip thumbnails generated in a Fusion Studio bin. This is an outdated path map since bins include now include the Studio Player.

**UserPaths:** Used for locations of studio- or facility-specific tools, like custom plugins and scripts located on a central server.

- **User Path Maps:** User paths are new paths that you have defined that do not currently exist in the Defaults settings.

Comp refers to the folder where the current composition is saved. For instance, saving media folders within the same folder as your Fusion Studio comp file is a way to use relative file paths for Loaders instead of actual file paths.

## Modifying a System Path Map

To modify an existing System path map, select the path map in the System section. Click the folder icon at the bottom of the Preferences window, and enter the name of the path map in the From field below. Enter the value of the path map in the To: field.

## Modifying a Default Path Map

To modify an existing Default path map, select the path map in the Default section. Click the folder icon at the bottom of the Preferences window, and enter the name of the path map in the From field below. Enter the value of the path map in the To: field.

## Creating a User Path Map

To create a path map, click on the New button and enter the name of the path map in the From field below. Enter the value of the path map in the To: field.

## Deleting a Path Map

To delete a user-created path map, select it from the list and click the Delete button. System and Default path maps cannot be deleted; only user created path maps can be removed from the Path Maps list.

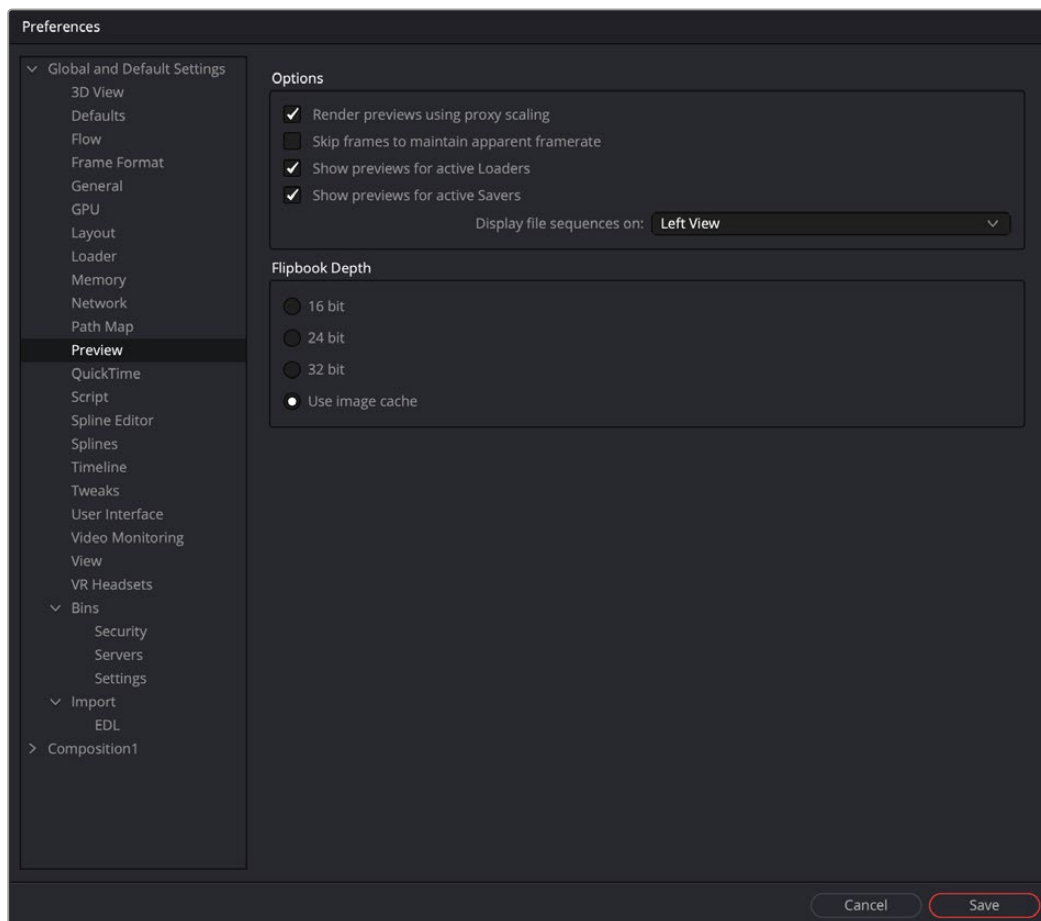
## Nesting Path Maps

When defining your own path map, you can use an existing path map in the new definition. For example, define a path map called 'Episode' that maps to MyDrive\ Projects\Episode1. Then create new path maps called Renders and Stills that map to Episode\ Renders\_v1 and Episode\Stills\_v1.

## Preview

Preview is only available in Fusion Studio. Previews in DaVinci Resolve use the Scratch Disk setting in the Media Storage preferences.

In the Preview preferences, you configure the creation and playback options for preview renders.



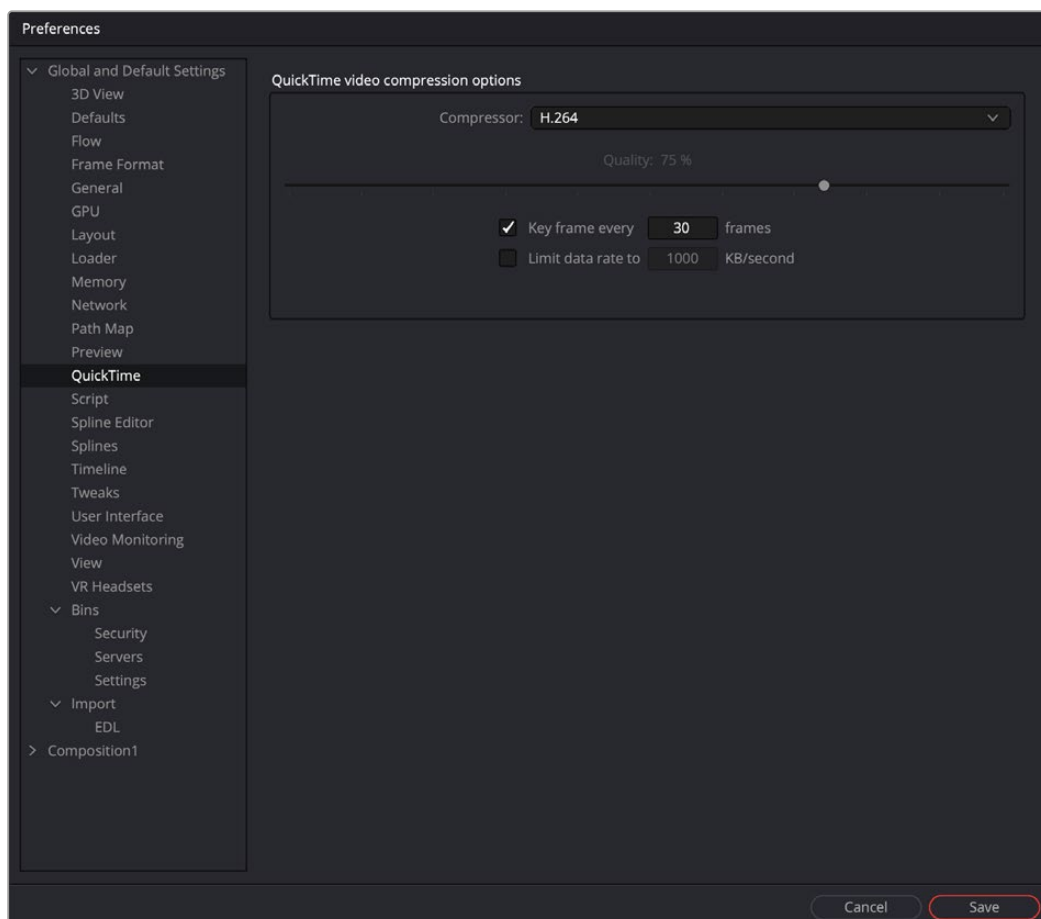
The Preview preferences

## Options

- **Render Previews Using Proxy Scaling:** When checked, this option scales down the images to the preview size for the Loader and Creator tools. This causes much faster rendering. If this option is disabled, frames will be rendered at full size and are then scaled down.
- **Skip Frames to Maintain Apparent Framerate:** When checked, frames are skipped during playback of Flipbooks and file sequences to maintain the frame rate setting.
- **Show Previews for Active Loaders:** This setting determines whether the preview playback controls are shown below the Inspector when a Loader with a valid file is activated.
- **Show Previews for Active Savers:** This setting determines whether the preview playback controls below the Inspector are shown when a Saver with a valid file is activated.
- **Display File Sequences On:** This setting determines which viewer or external monitor is used for the interactive and file sequence playbacks as well as for the scrubbing function in the bins.

## QuickTime

The QuickTime preferences are only available in Fusion Studio on macOS. These settings configure the default QuickTime codec settings when you select QuickTime as the rendering file format in the Saver node.



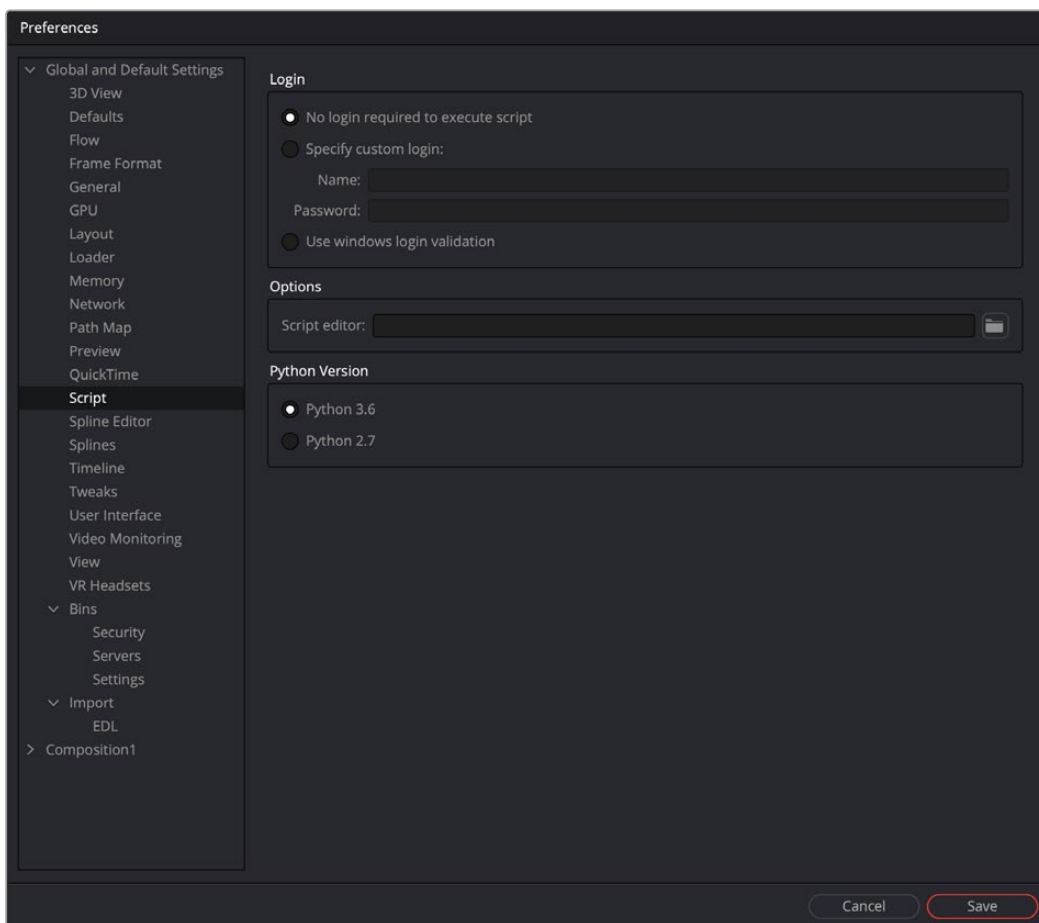
The QuickTime preferences

- **Compressor:** This drop-down menu displays the QuickTime codecs available from your computer. Fusion tests each codec when the program is started; therefore, some codecs may not be available if the tests indicate that they are unsuitable for use within Fusion.

- **Quality:** This slider is used to determine the amount of compression to be used by the codec. Higher values produce clearer images but larger files. Not all codecs support the Quality setting.
- **Key Frame Every X Frames:** When checked, the codec will create key frames at specified intervals. Key frames are not compressed in conjunction with previous frames and are, therefore, quicker to seek within the resulting movie. Not all codecs support the key frame setting.
- **Limit Data Rate To X KB/Second:** When checked, the data rates of the rendered file will be limited to the amount specified. Not all codecs support this option. Enter the data rate used to limit the QuickTime in kilobytes (kB) per second, if applicable. This control will have no effect if the Limit Data Rate To option is not selected.

## Script

The preferences for Scripting include a field for passwords used to execute scripts from the command line and applications for use when editing scripts.



The Script preferences

## Login

There are three login options for running scripts outside of the Fusion application.

- **No Login Required to Execute Script:** When enabled, scripts executed from the command line, or scripts that attempt to control remote copies of Fusion, do not need to log in to the workstation in order to run.

- **Specify Custom Login:** If a username and password are assigned, Fusion will refuse to process incoming external script commands (from FusionScript, for example), unless the Script first logs in to the workstation. This only affects scripts that are executed from the command line, or scripts that attempt to control remote copies of Fusion. Scripts executed from within the interface do not need to log in regardless of this setting. For more information, see the Scripting documentation.
- **Use Windows Login Validation:** When using Fusion on Windows, enabling this option verifies the user name and password (also known as credentials) with the operating system before running the script.

## Options

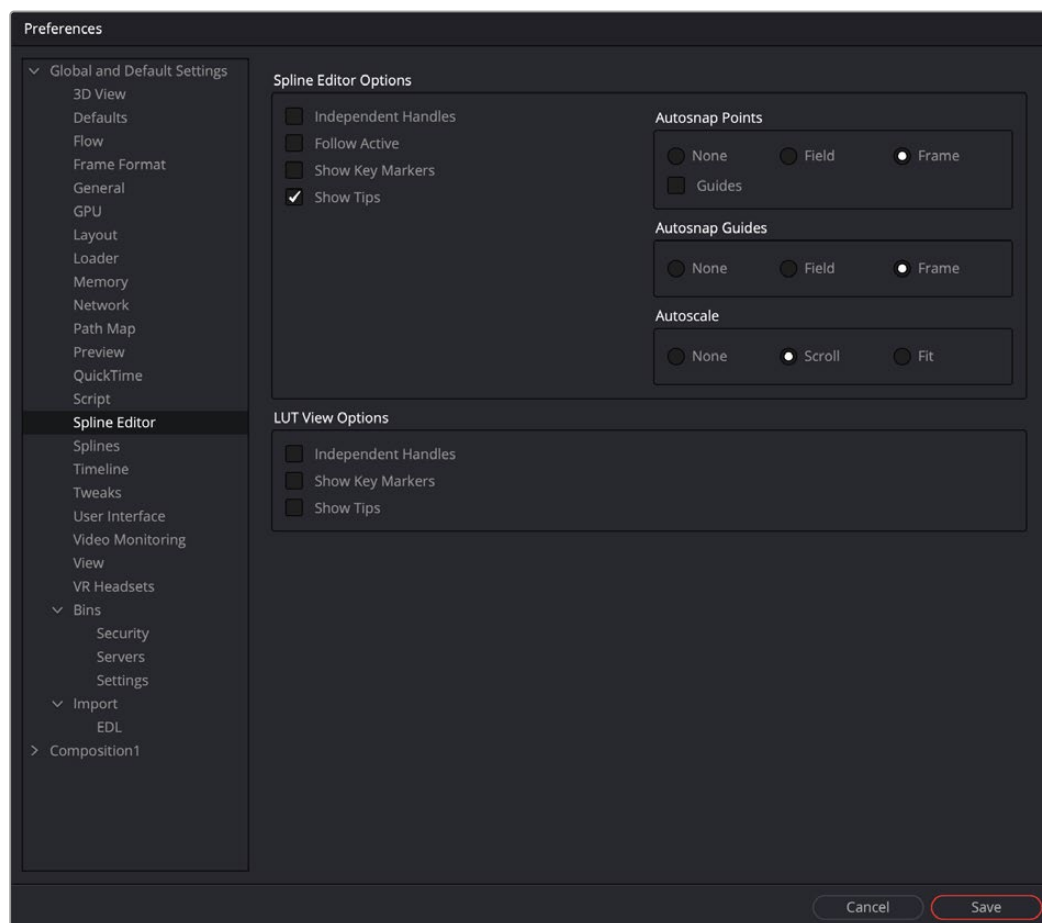
**Script Editor:** Use this preference to select an external editor for scripts. This preference is used when selecting Scripts > Edit.

## Python Version

Two options are presented here for selecting the version of Python that you plan on using for your scripts.

## Spline Editor

The Spline Editor preferences allow you to set various spline options for Autosnap behavior, handles, markers, and more. This only affects splines displayed in the Spline Editor, not splines created in the viewer using the polygon tool or paths.



The Spline Editor preferences

## Spline Editor Options

These settings control the spline behavior in the Spline Editor, as well as the appearance of the graph area.

- **Independent Handles:** Enabling this option allows the In or Out direction handle on newly created key frames to be moved independently without affecting the other. This option is also available via the Options submenu when right-clicking in the Spline Editor graph.
- **Follow Active:** The Spline Editor focuses on the currently active tool. This option is also available via the Options submenu when right-clicking in the Spline Editor graph.
- **Show Key Markers:** Small colored triangles will be displayed at the top of the Spline Editor Time Ruler to indicate key frames on active splines. The colors of the triangles match the colors of the splines. This option is also available via the Show submenu when right-clicking in the Spline Editor graph.
- **Show Tips:** Toggles if tooltips are displayed or not. This option is also available via the Show submenu when right-clicking in the Spline Editor graph.
- **Autosnap Points:** When moving points in the Spline Editor, these will snap to the fields or frames or can be moved freely. This option is also available via the Options submenu when right-clicking in the Spline Editor graph.
- **Guides:** When moving points in the Spline Editor, these will snap to guides as well. This option is also available via the Options submenu when right-clicking in the Spline Editor graph.
- **Autosnap Guides:** When moving or creating guides, these will snap to the fields or frames or can be moved freely. This option is also available via the Options submenu when right-clicking in the Spline Editor graph.
- **Autoscale:** Keeps the Spline Editor scales intact on changing the editable spline content of the graph. This scale is also available via the Options submenu when right-clicking in the Spline Editor graph.
- **Scroll:** Scrolls horizontally and vertically to show all or most of the spline points. This option is also available via the Scale submenu when right-clicking in the Spline Editor graph.
- **Fit:** Zooms to fit all points within the spline graph, if necessary. This option is also available via the Scale submenu when right-clicking in the Spline Editor graph.

## LUT View Options

These settings let you control how the LUT view is displayed.

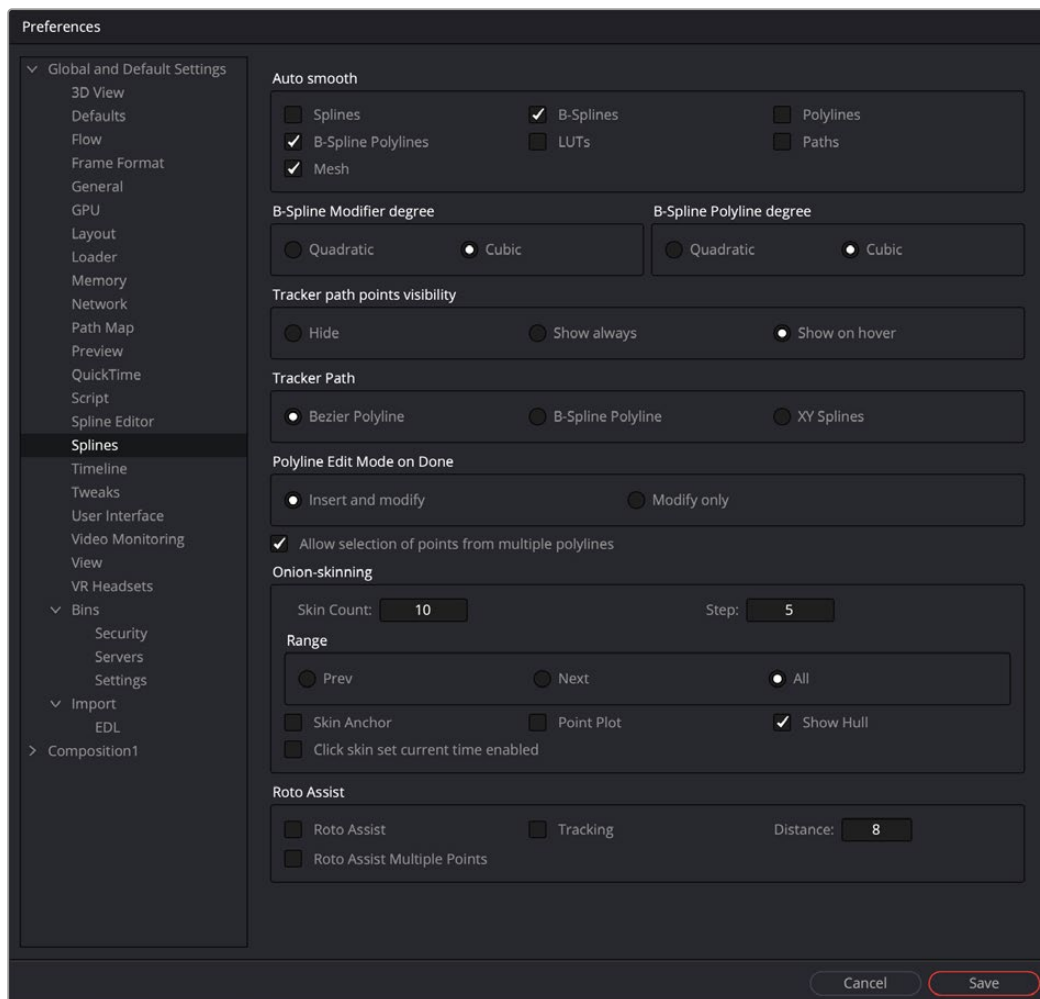
- **Independent Handles:** Enabling this option allows the In or Out direction handle on newly created key frames to be moved independently without affecting the other.
- **Show Key Markers:** Small colored triangles will be displayed at the top of the Spline Editor Time Ruler to indicate key frames on active splines. The colors of the triangles match the colors of the splines.
- **Show Tips:** Toggles whether tooltips are displayed.

## Splines

Options for the handling and smoothing of animation splines, tracker path defaults, and rotoscoping are found in the Splines preferences.

- **Autosmooth:** Automatically smooths out any newly created points or key frames on the splines selected in this section. You can choose to automatically smooth animation splines, B-Splines, polyline matte shapes, LUTs, paths, and meshes.
- **B-Spline Modifier Degree:** This setting determines the degree to which the line segments influence the resulting curvature when B-Splines are used in animation. Cubic B-Splines determine a segment through two control points between the anchor points, and Quadratic B-Splines determine a segment through one control point between the anchor points.

- **B-Spline Polyline Degree:** This setting is like the one above but applies to B-Splines used for masks.

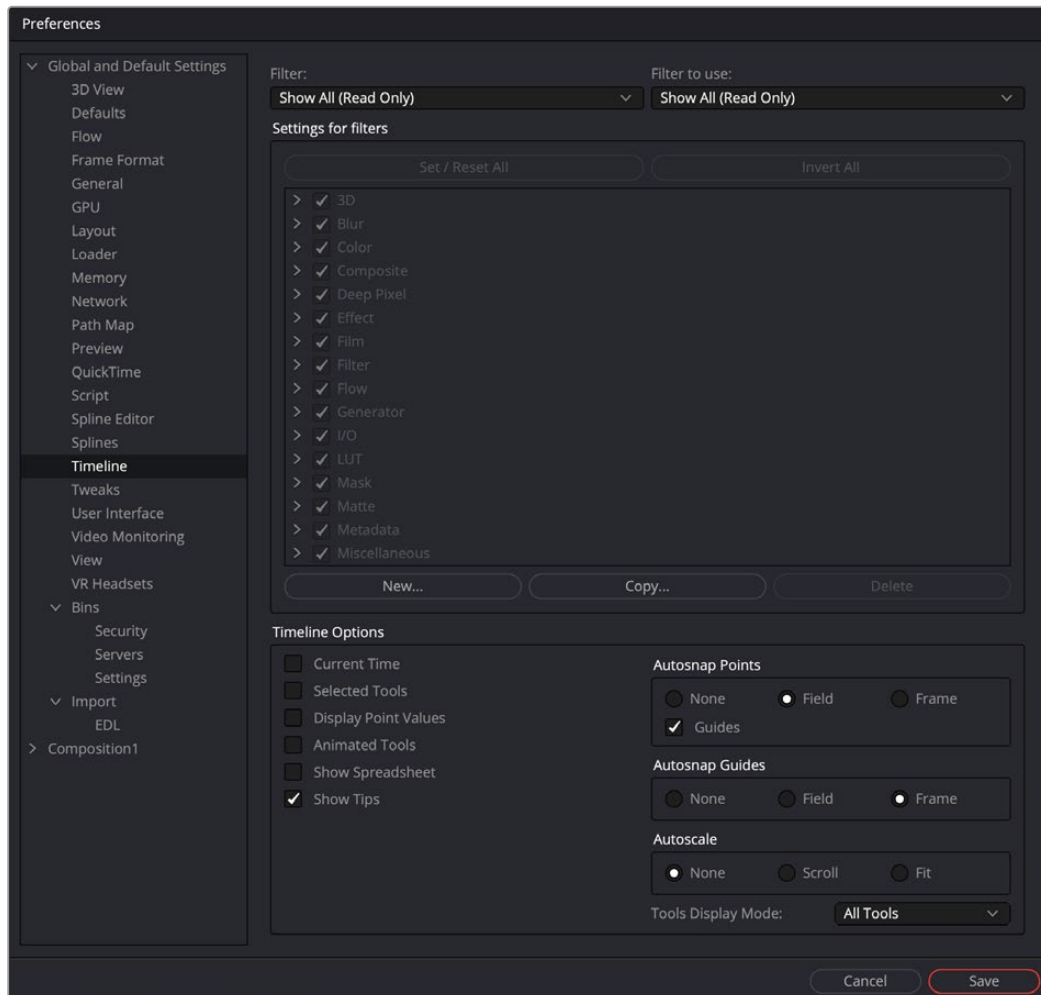


The Splines preferences

- **Tracker Path Points Visibility:** This setting determines the visibility of the control points on tracker paths. You can show them, hide them, or show them when your cursor hovers over the path, which is the default behavior.
- **Tracker Path:** The default tracker creates Bézier-style spline paths. Two other options in this setting allow you to choose B-Spline or XY Spline paths.
- **Polyline Edit Mode on Done:** This setting determines the state of the Polyline tool after you complete the drawing of a polyline. It can either be set to modify the existing control points on the spline or modify and add new control points to the spline.
- **Onion Skinning:** The Onion Skinning settings determine the number of frames displayed while rotoscoping, allowing you to preview and compare a range of frames. You can also adjust if the preview frames only from the frame prior to the current frame, after the current frames, or split between the two.

## Timeline

The Timeline preferences is where you create and edit Keyframes Editor/Spline Editor filters and set default options for the Keyframes Editor.



The Timeline preferences

### Filter/Filter to Use

The Filter menu populates the hierarchy area below the menu with that setting. It lets you edit the filters. The Filter to Use menu selects the default filter setting located in the Keyframes Editor Options menu.

### Settings for Filters

This area is used to create a new filter and define its settings. You start by first clicking the New button and entering the name of the new Filter. You then select any of the tools that you want the filter to contain. Only tools that are checked will appear in the Keyframes Editor or Spline Editor when the filter is selected. You can also create a copy of the filter using the Copy button or remove a filter from the list by clicking the Delete button.

### Timeline Options

The Timeline Options configure which options in the Keyframe Editor are enabled by default. A series of checkboxes correspond to buttons located in the Timeline, allowing you to determine the states of those buttons at the time a new comp is created.

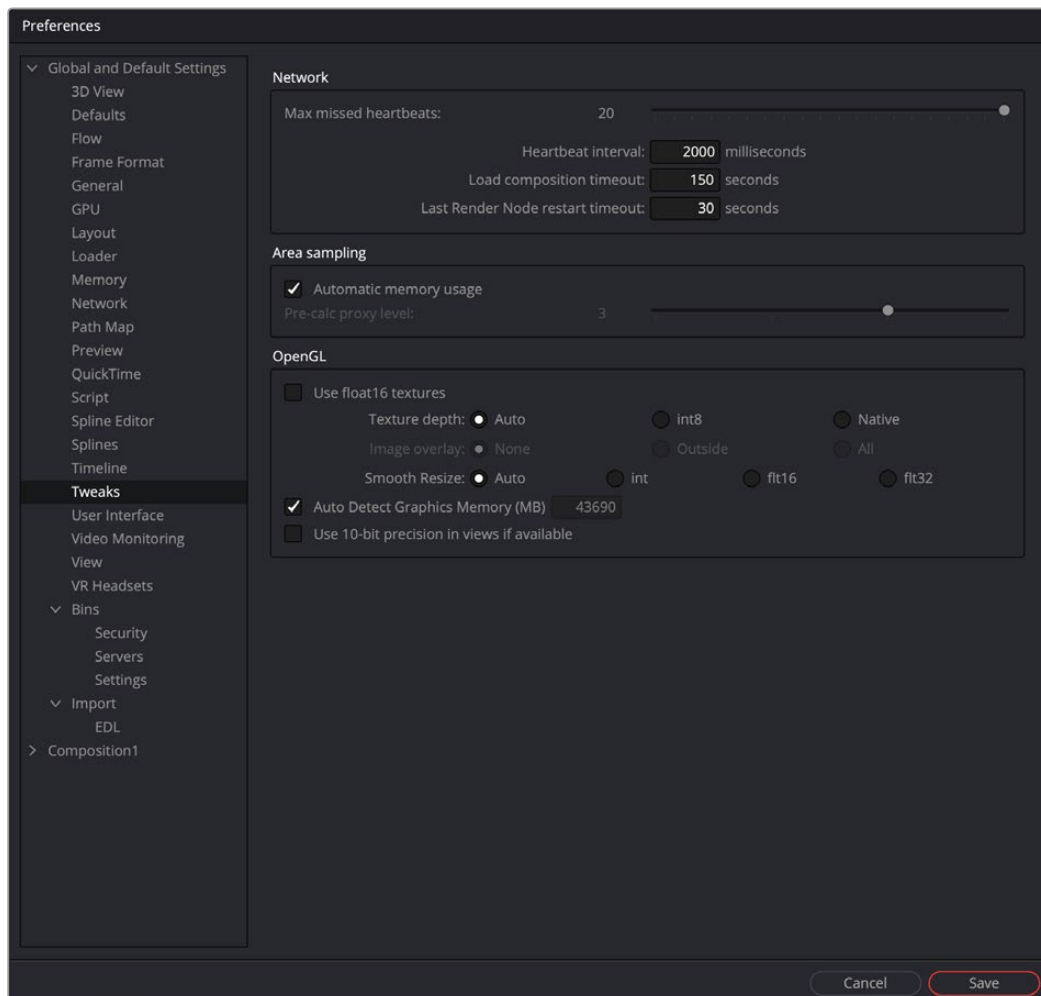


For more information on the Keyframes Editor functions, see *Chapter 71, “Animating in Fusion’s Keyframes Editor,”* in the DaVinci Resolve Reference Manual or Chapter 9 in the Fusion Reference Manual.

- **Autosnap Points:** When moving points in the Keyframes Editor, the points will snap to the fields or to the frames, or they can be moved freely.
- **Guides:** When moving points in the Keyframes Editor, the point will snap to the guides that are placed in the Timeline graph.
- **Autosnap Guides:** When moving or creating guides, the guides will snap to the fields or to the frames, or they can be moved freely.
- **Autoscale:** Keeps the Timeline scales intact while changing the editable spline content in the graph. When set to scroll, the Timeline scrolls horizontally and vertically to show all or most of the spline points when changing the editable spline content in the graph. When set to Fit, the Timeline zooms to fit all points within the graph, if necessary.
- **Tools Display Mode:** This menu controls the default sort order of the tools displayed in the Keyframes Editor. The default can be changed using the Sort order menu in the upper right of the Keyframes Editor.

## Tweaks

The Tweaks preferences handle a collection of settings for fine-tuning Network rendering in Fusion Studio and graphics hardware behavior.



The Tweaks preferences

## Network

The Network section is used to control and monitor the health of communication packets over TCP/IP when rendering over a network in Fusion Studio.

- **Maximum Missed Heartbeats:** This setting determines the maximum number of times the network is checked before terminating the communication with a Render node.
- **Heartbeat Interval:** This sets the time between network checks.
- **Load Composition Timeout:** This timeout option determines how long the Render Manager will wait for a composition to load before moving on to another task.
- **Last Slave Restart Timeout:** This timeout option determines how long the Render Manager will wait for a render slave to respond before using another render slave.

## File I/O

The File I/O options are used to control the performance when reading frames or large media files from both direct and networked attached storage.

- **I/O Canceling:** This option enables a feature of the operating system that allows queued operations to be canceled when the function that requested them is stopped. This can improve the responsiveness, particularly when loading large images over a network.

Enabling this option will specifically affect performance while loading and accessing formats that perform a large amount of seeking, such as the TIFF format.

This option has not been tested with every hardware and OS configuration, so it is recommended to enable it only after you have thoroughly tested your hardware and OS configuration using drive loads from both local disks and network shares.

- **Enable Direct Reads:** Enabling this checkbox uses a more efficient method when loading a large chunk of contiguous data into memory by reducing I/O operations. Not every operating system employs this ability, so it may produce unknown behavior.
- **Read Ahead Buffers:** This slider determines the number of 64K buffers that are used to read ahead in a file I/O operation. The more buffers, the more efficient loading frames from disk will be, but the less responsive it will be to changes that require disk access interactively.

## Area Sampling

The Area Sampling options allow you to fine-tune the RAM usage on Render nodes by trading off speed for lower RAM requirements.

- **Automatic Memory Usage:** This checkbox determines how area sampling uses available memory. Area sampling is used for Merges and Transforms. When the checkbox is enabled (default), Fusion will detect available RAM when processing the tool and determine the appropriate trade-off between speed and memory.

If less RAM is available, Fusion will use a higher proxy level internally and take longer to render. The quality of the image is not compromised in any way, just the amount of time it takes to render. In node trees that deal with images larger than 4K, it may be desirable to override the automatic scaling and fix the proxy scale manually. This can preserve RAM for future operations.

- **Pre-Calc Proxy Level:** Deselecting the Automatic Memory will enable the Pre-Calc Proxy Scale slider. Higher values will use less RAM but take much longer to render.

## Open GL

This section controls how Fusion makes use of your graphics card when compositing in 3D with the Renderer 3D node. Most settings may be left as they are, but since OpenGL hardware varies widely in capabilities and different driver revisions can sometimes introduce bugs, these tweaks can be useful if you are experiencing unwanted behavior.

- **Disable View LUT Shaders:** OpenGL shaders can often dramatically accelerate View LUTs, but this can occasionally involve small trade-offs in accuracy. This setting will force Fusion to process LUTs at full accuracy using the CPU instead. Try activating this if View LUTs do not seem to be giving the desired result.
- **Use Float16 Textures:** If your graphics hardware supports 16-bit floating-point textures, activating this option will force int16 and float32 images to be uploaded to the viewer as float16 instead, which may improve playback performance.
- **Texture Depth:** Defines in what depth images are uploaded to the viewer.

**Auto:** The Auto option (recommended) lets Fusion choose the best balance of performance and capability.

**int8:** Similar to the Use Float16 Textures switch, this option can be used to force images to be uploaded to the Display View as int8, which can be faster but gives less range for View LUT correction.

**Native:** The Native option uploads images at their native depth, so no conversion is done.

- **Image Overlay:** The Image Overlay is a viewer control used with Merge and Transform tools to display a translucent overlay of the transformed image. This can be helpful in visualizing the transformation when it is outside the image bounds but may reduce performance when selecting the tool if cache memory is low. There are three settings to choose from: None, Outside, and All.

**None:** This setting never displays the translucent overlay or controls, which can reduce the need for background renders, in some cases resulting in a speed up of the display.

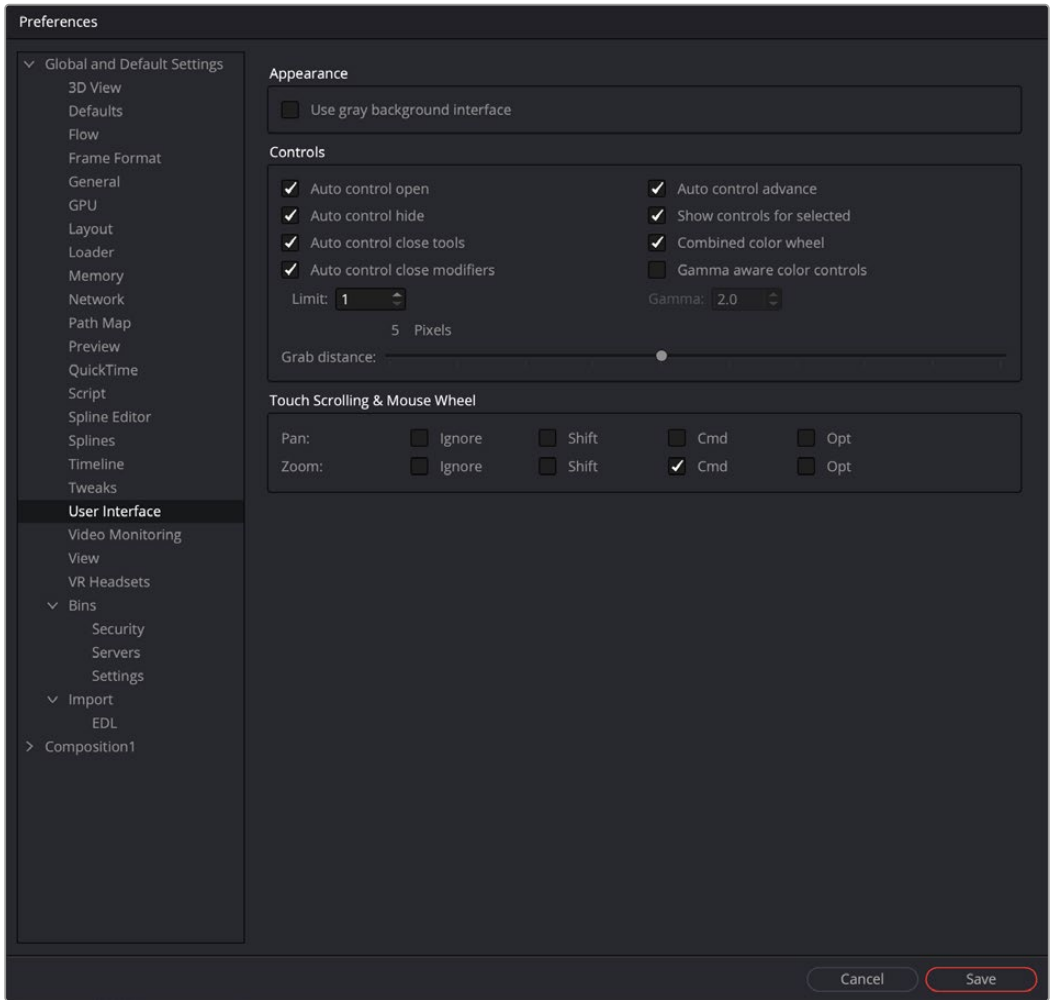
**Outside:** This will display only those areas of the control that are outside the bounds of the image, which can reduce visual confusion.

**All:** Displays all overlays of all selected tools.

- **Smooth Resize:** This setting can disable the viewer's Smooth Resize behavior when displaying floating-point images. Some older graphics cards are not capable of filtering floating-point textures or may be very slow. If Smooth Resize does not work well with float images, try setting this to flt16 or int.
- **Auto Detect Graphics Memory (MB):** Having Fusion open alongside other OpenGL programs like 3D animation software can lead to a shortage of graphics memory. In those cases, you can manually reduce the amount of memory Fusion is allowed to use on the card. Setting this too low or too high may cause performance or data loss.
- **Use 10-10-10-2 Framebuffer:** If your graphics hardware and monitor support 30-bit color (Nvidia Quadro/AMD Radeon Pro, and some Nvidia GeForce/AMD Radeon), this setting will render viewers with 10 bits per primary accuracy, instead of 8 bits. Banding is greatly reduced when displaying 3D renders or images deeper than 8-bit.

# User Interface

The User Interface preferences set the appearance of the user interface window and how the Inspector is displayed.



The User Interface preferences

## Appearance

When enabled, the Use Gray Background Interface checkbox will change the color of the background in Fusion’s panels to a lighter, more neutral shade of gray.

## Controls

This group of checkboxes manages how the controls in the Inspector are displayed.

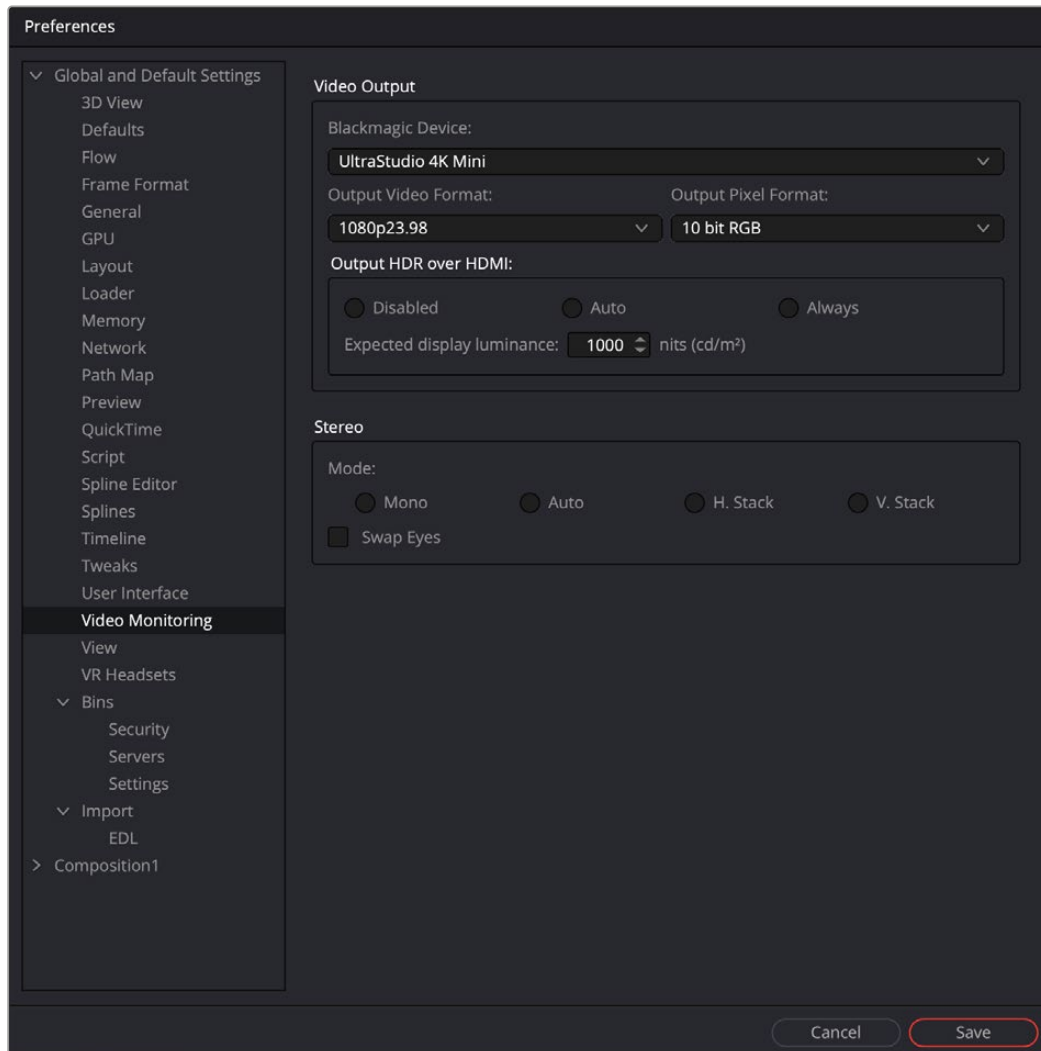
- **Auto Control Open:** When disabled, only the header of the selected node is displayed in the Inspector. You must double-click the header to display the parameters. When enabled, the parameters are automatically displayed when the node is selected.
- **Auto Control Hide:** When enabled, only the parameters for the currently active tool (red outline) will be made visible. Otherwise, all tool headers will be visible and displayed based on the Auto Control Open setting.
- **Auto Control Close Tools:** When enabled, only the active (red outlined) tool in the Node Editor will have controls displayed. Any previous active node's tools will be closed in the Inspector. When disabled, any number of tools may be opened to display parameters at the same time. This setting has no effect if the Auto Control Hide checkbox is enabled.
- **Auto Control Close Modifiers:** When enabled, only one modifier's parameters will be displayed for the active node. Any additional modifiers for the active node will show only their header.
- **Auto Control Advance:** If the Auto Control Advanced checkbox is enabled, the Tab key and Return/Enter key will cause the keyboard focus to advance to the next edit box within the Inspector. When disabled, Return/Enter will cause the value entered to be accepted, but the keyboard focus will remain in the same edit box of the control. The Tab key can still be used to advance the keyboard focus.
- **Show Controls for Selected:** When this option is disabled, only the active tool's parameters are shown in the Inspector. By default, it is enabled, showing controls for the active tool as well as all selected tools.
- **Combined Color Wheel:** When the Color Corrector tool is displayed in the Inspector, enabling this checkbox will show one color wheel with buttons to switch between the master, shadow, midtones, and highlight channels. Otherwise, four color wheels are displayed in the Inspector.
- **Gamma Aware Color Controls:** This setting adjusts color correction nodes when working with Rec. 709 images in a non-color managed project. Rec. 709 images appear correct on the computer monitor because monitors have a gamma adjustment built in. When working in the Rec. 709 color space without color management, enabling Gamma Aware color removes the gamma, applies the color correction as if it were linear, and then reapplies the gamma. For Rec. 709 images, enable the Gamma Aware setting and enter a Gamma value of 2.4. In a color managed linear project, this should be set to Off or a value of 1.0. When dealing with mixed color spaces, Fusion reads the metadata from the image and sets the Aware gamma value based on the metadata available.
- **Grab Distance:** This slider ranges from 1 to 10 and defaults to 5. It designates the active area around the mouse pointer and can be modified if you have difficulties in selecting points for modification in paths and spline curves. Smaller values will require a more accurate selection with the mouse pointer.

## Touch Scrolling and Mouse Wheel

This group of settings allows you to configure which, if any, keyboard modifiers are needed to pan or zoom a panel when using a trackpad or middle mouse wheel.

## Video Monitoring

This setting is only available in Fusion Studio. Control over video hardware for the Fusion Page is done in the DaVinci Resolve preferences. The Video Monitoring preferences are used to configure the settings of Blackmagic Design capture and playback products such as DeckLink PCIe cards and UltraStudio i/O units.



The Video Monitoring preferences

### Video Output

This group of drop-down menus allows you to select the type of video I/O device you have installed, the output resolution, and the pixel format. These settings have nothing to do with your rendered output; it is only for your display hardware.

The Output HDR over HDMI settings are used to output the necessary metadata when sending high dynamic range signals over HDMI 2.0a and have it correctly decoded by an HDR capable video display.

The Auto setting detects the image's values and outputs HDR. This will not affect non HDR images.

The Always setting forces HDR on all the time. This can be useful when checking non HDR and HDR grades.

When Auto or Always is selected, you can then set the “nit” level (slang for  $\text{cd/m}^2$ ) to whatever peak luminance level your HDMI connected HDR display is capable of.

## Stereo Mode

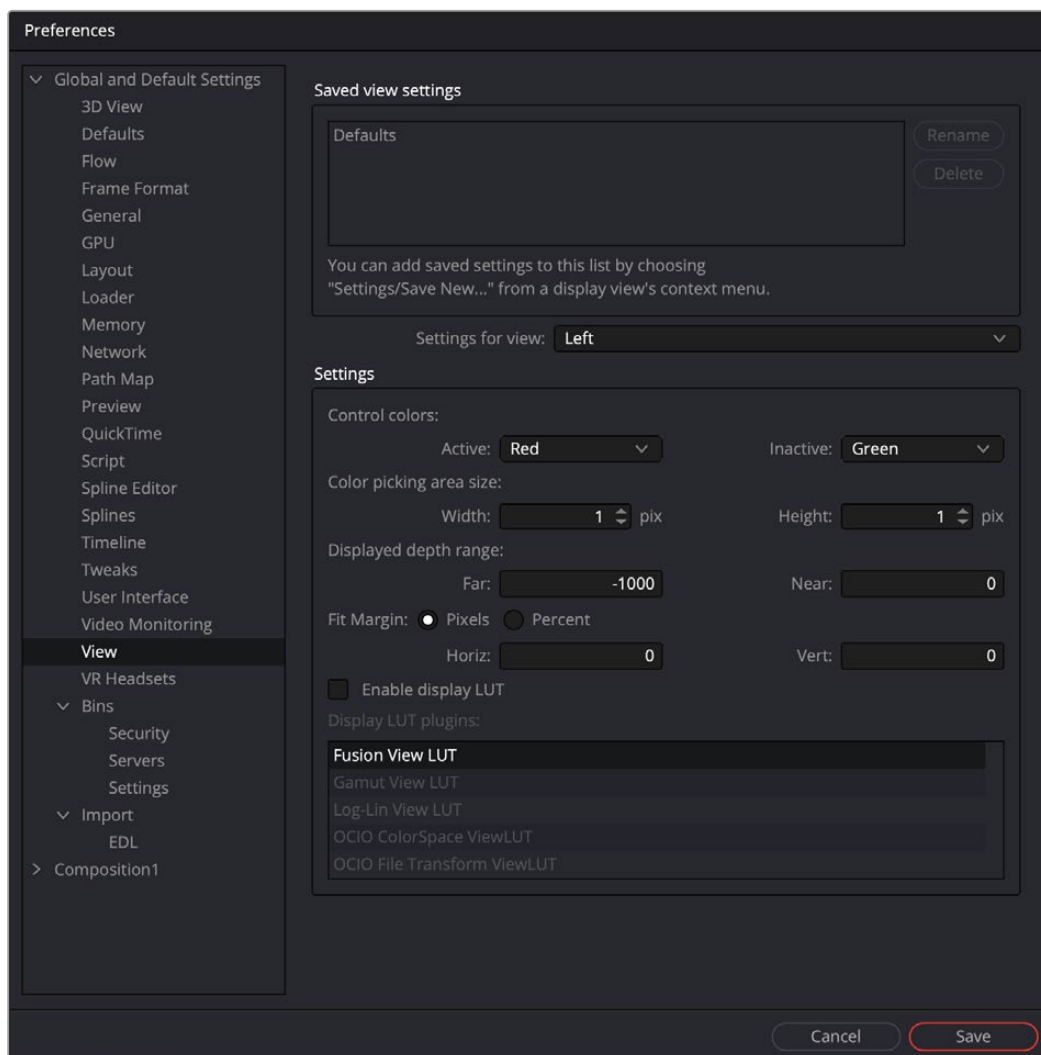
This group of settings configures the output hardware for displaying stereo 3D content.

- Mono will output a single non stereo eye.
- Auto will detect which method with which the stereo images are stacked.
- Use the Vstack option if the stereo images are stacked vertically as left on top and right at the bottom.
- Use the Hstack option if the stereo images are stacked horizontally as left and right.

The Swap eyes checkbox will swap the eyes if stereo is reversed.

## View

The View preferences are used to manage settings and default controls for viewers.



The View preferences

## Saved View Settings

The area at the top of the view preferences lists the currently saved settings that you create from the viewer's contextual menu. You can use the Rename and Delete buttons to manage the selected entries in the list.

For more information on the viewer and its contextual menu, see *Chapter 69, "Using Viewers."* in the DaVinci Resolve Reference Manual or Chapter 7 in the Fusion Reference Manual.

## Settings for View

Each viewer has its own preferences. The Settings for View drop-down menu is used to select the viewer you want to configure.

## Control Colors

The Control Colors setting allows you to determine the color of the active/inactive onscreen controls.

## Color Picking Area Size

You can use these width/height controls to set the number of pixels sampled when using the Color Picker in the viewers.

## Displayed Depth Range

The Displayed Depth Range setting controls the view normalization of the Z-Channel.

## Fit Margin

The Fit Margin setting determines how much padding is left around the frame when the Fit button is pressed or Fit is selected from the viewer's contextual menu.

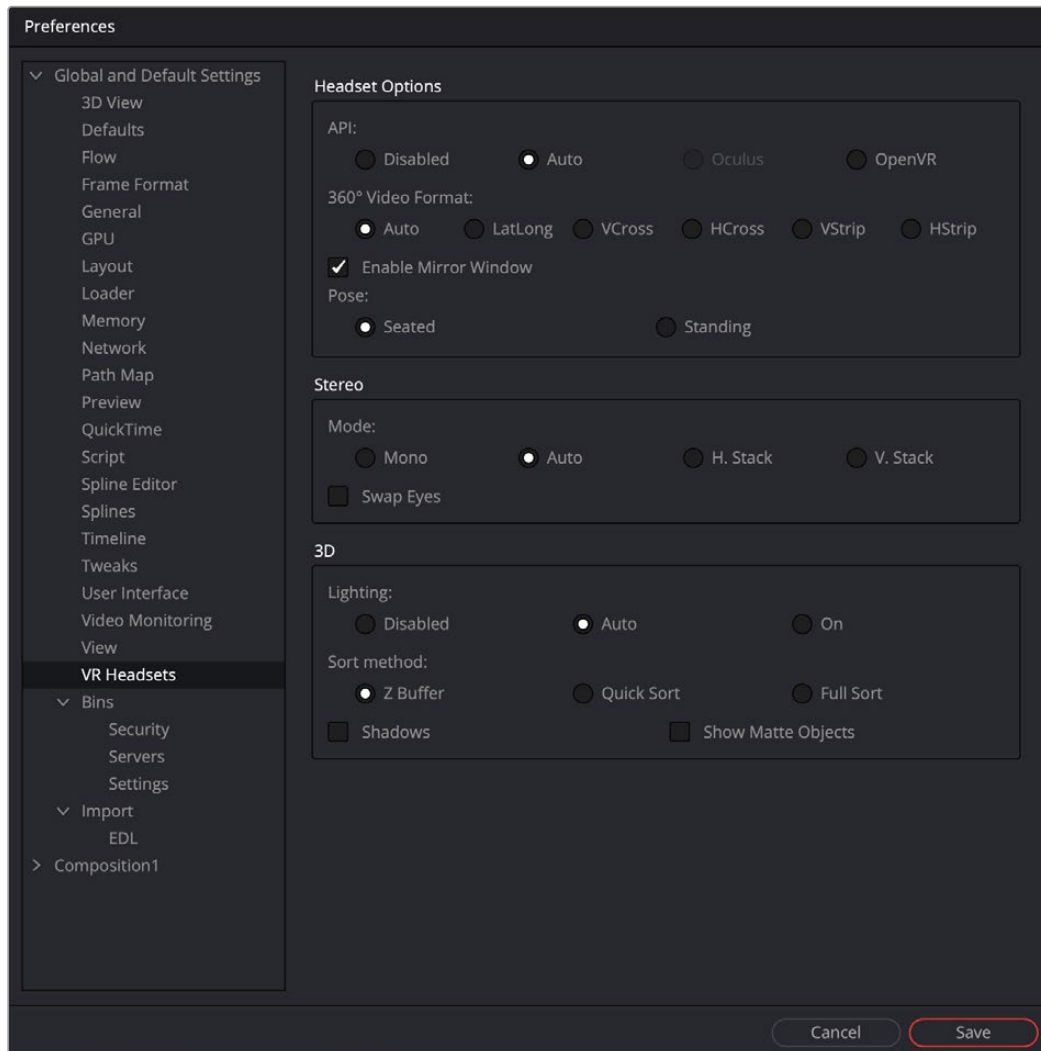
- **Display LUT Plugins**

This list shows the available display LUTs and activates the selected one as default.



## VR Headsets

The VR Headsets preferences allow configuration of any connected Virtual Reality headsets, including how stereo and 3D scenes are viewed.



The VR Headsets preferences

### Headset Options

The Headset options are used to select the type of VR headset you are using to view the composite as well as the video layout of the 360° view.

#### API

- **Disabled:** Disabled turns off and hides all usage of headsets.
- **Auto:** Auto will detect which headset is plugged in.
- **Oculus:** Oculus will set the VR output to the Oculus headset.
- **OpenVR:** OpenVR will support a number of VR headsets like the HTC Vive.

### 360° Video Format

- **Auto:** Auto will detect the incoming image layout from the metadata and image frame aspect.
- **VCross and HCross:** VCross and HCross are the six square faces of a cube laid out in a cross, vertical or horizontal, with the forward view in the center of the cross, in a 3:4 or 4:3 image.
- **VStrip and HStrip:** VStrip and HStrip are the six square faces of a cube laid vertically or horizontally in a line, ordered as Left, Right, Up, Down, Back, Front (+X, -X, +Y, -Y, +Z, -Z), in a 1:6 or 6:1 image.
- **LatLong:** LatLong is a single 2:1 image in equirectangular mapping.
- **Enable Mirror Window:** Enable Mirror Window will show a window displaying the headset user's live view.

## Stereo

Similar to normal viewer options for stereo 3D comps, these preferences control how a stereo 3D comp is displayed in a VR headset.

### Mode

- **Mono:** Mono will output a single non stereo eye.
- **Auto:** Auto will detect the method with which the stereo images are stacked.
- **Vstack:** Vstack stereo images are stacked vertically as left on top and right at the bottom.
- **Hstack:** Hstack stereo images are stacked horizontally as left and right.
- **Swap Eyes:** Swap eyes will swap the eyes if stereo is reversed.

## 3D

Similar to normal viewer options for 3D comps, these preferences control how a 3D comp is displayed in a VR headset.

### Light

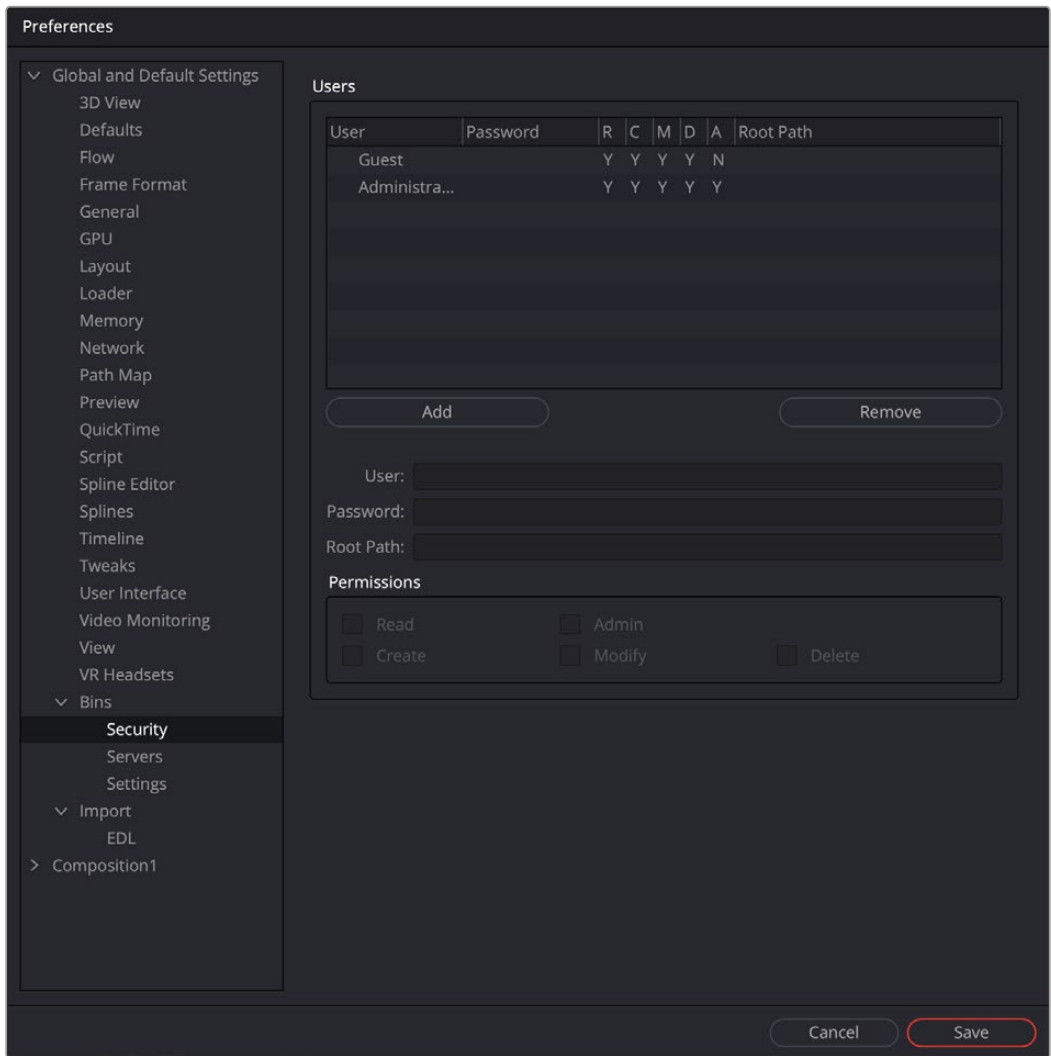
- Disabled lighting is off.
- Auto will detect if lighting is on in the view.
- On will force lighting on in the VR view.

### Sort Method

- Z buffer sorting is the fast OpenGL method of sorting polygons.
- Quick Sort will sort the depth of polygons to get better transparency rendering.
- Full Sort will use a robust sort and render method to render transparency .
- Shadows can be on or off.
- Show Matte Objects will make matte objects visible in view or invisible.

## Bins/Security

Bins preferences are only available in Fusion Studio. These preferences are used to manage the Bin users and their permissions.



The Bins Security preferences

### Users List

The Users List is a list of the users and their permissions. You can select one of the entries to edit their settings using the User and Password edit boxes.

- **Add:** The Add button is used to add a new user to the list by entering a username and password.
- **Remove:** Click this button to remove the selected entry.

### User

This editable field shows the username for the selected Bin Server item. If the username is unknown, try “Guest” with no password.

### Password

Use this field to enter the password for the Bin user entered in the Users list.

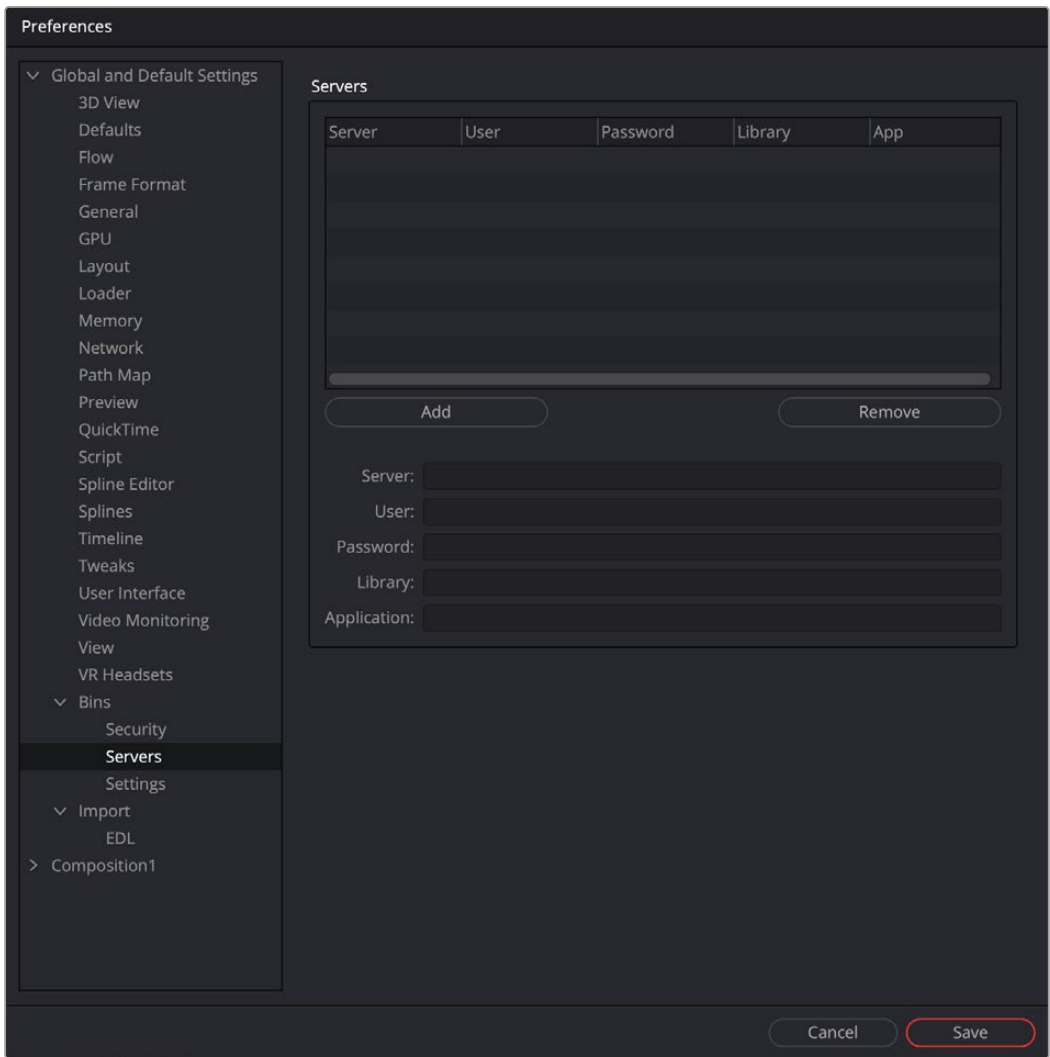
### Permissions

The administrator can set up different permission types for users.

- **Read:** This will allow the user to have read-only permission for the bins.
- **Create:** This will allow the user to create new bins.
- **Admin:** This gives the user full control over the bins system.
- **Modify:** This allows the user to modify existing bins.
- **Delete:** This allows the user to remove bins.

### Bins/Server

These preferences are used to add Bin Servers to the list of bins Fusion will display in the Bins dialog.



The Bin Servers preferences

### Servers

This dialog lists the servers that are currently in the connection list. You can select one of the entries to edit its settings.

- **Add:** Use this button to add a new server to the list.
- **Remove:** Click this button to remove the selected entry.

## Server

This editable field shows the name or IP address of the server for the selected entry in the list.

## User

This editable dialog shows the username for the selected Bin Server item.

## Password

Use this field to enter the password for the server entered in the Server list.

## Library

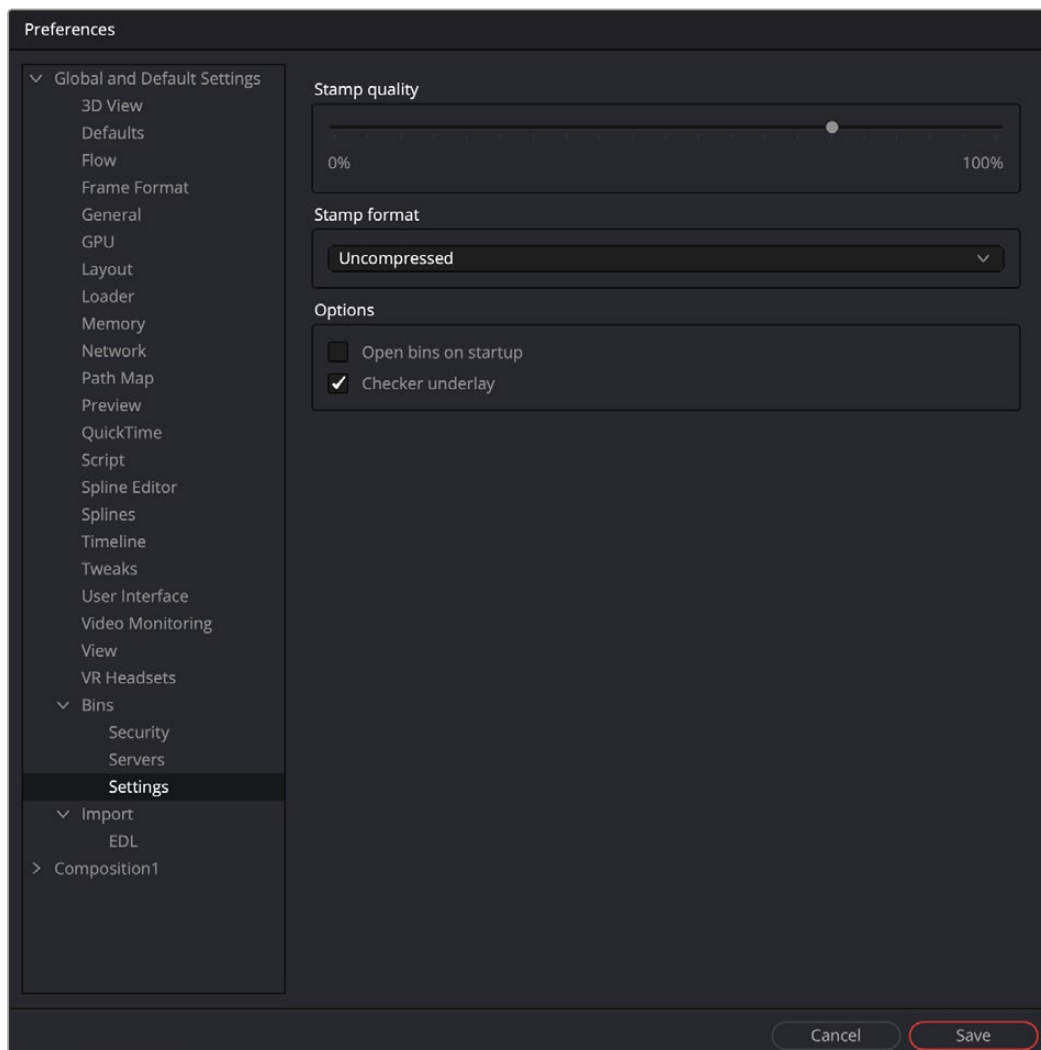
The Library field lets you name the bins. If you wanted to create a bin for individual projects, you would name it in the Library field and each project would get its own bin.

## Application

The Application field allows larger studios to specify some other program to serve out the Bin requests.

## Bins/Settings

These preferences are used to control the default behavior of bins.



The Bins Settings preferences

## Stamp Quality

The Stamp Quality is a percentage slider that determines the compression ratio used for Stamp thumbnail creation. Higher values offer better quality but take up more space.

## Stamp Format

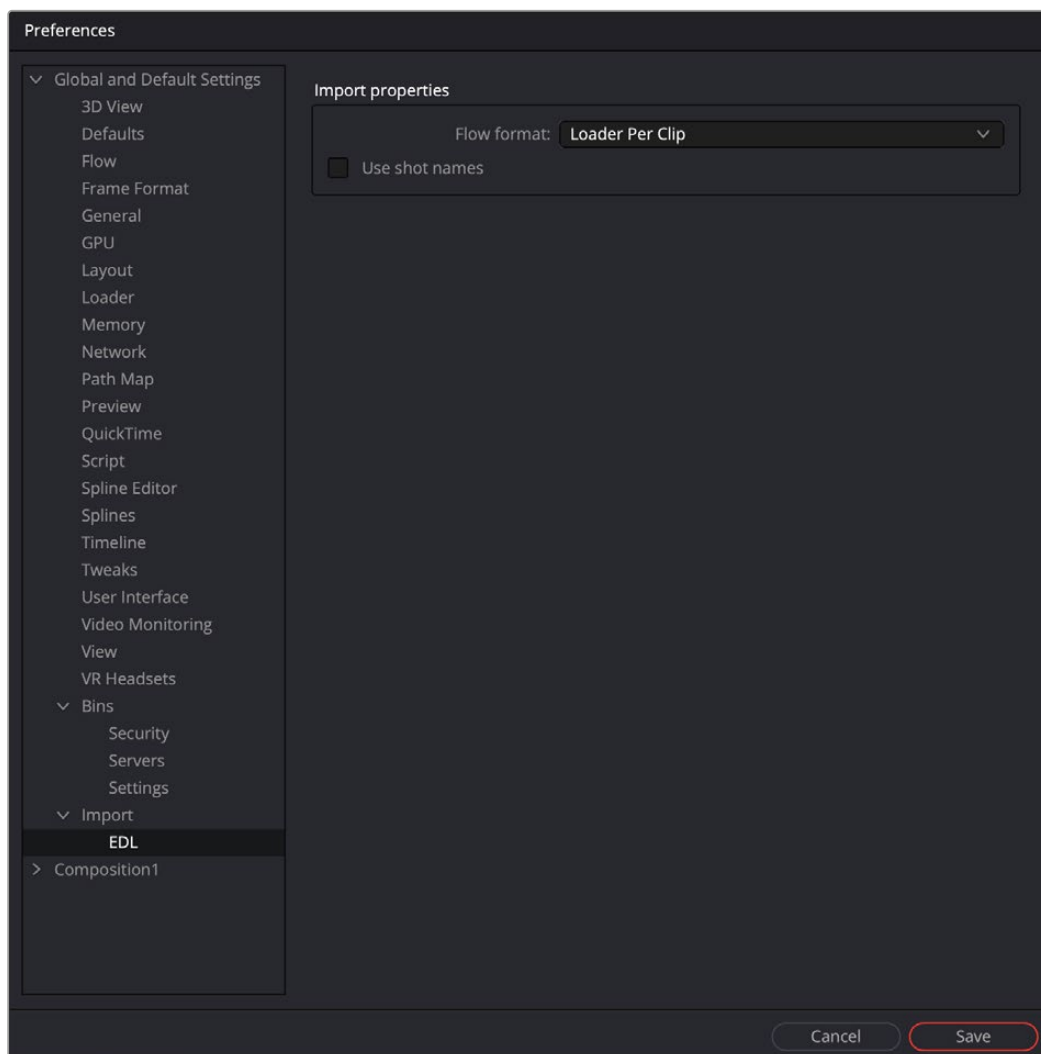
This drop-down list determines whether the Stamp thumbnails will be saved as compressed or uncompressed.

## Options

- **Open Bins on Startup:** When Open Bins on Startup is checked, the bins will open automatically when Fusion is launched.
- **Checker Underlay:** When the Checker Underlay is enabled, a checkerboard background is used for clips with alpha channels. When disabled, a gray background matching the Bin window is used as the clip's background.

## EDL Import

The EDL Import options are used to determine how compositions are created from imported CMX-formatted EDL files.



The EDL Import preferences

## Flow Format

This drop-down menu provides three options that determine how the node tree is constructed for the imported EDL file.

- **Loader Per Clip:** A Loader will be created for each clip in the EDL file.
- **A-B Roll:** A node tree with a Dissolve tool will be created automatically.
- **Loader Per Transition:** A Loader with a Clip list will be created, representing the imported EDL list.

## Use Shot Names

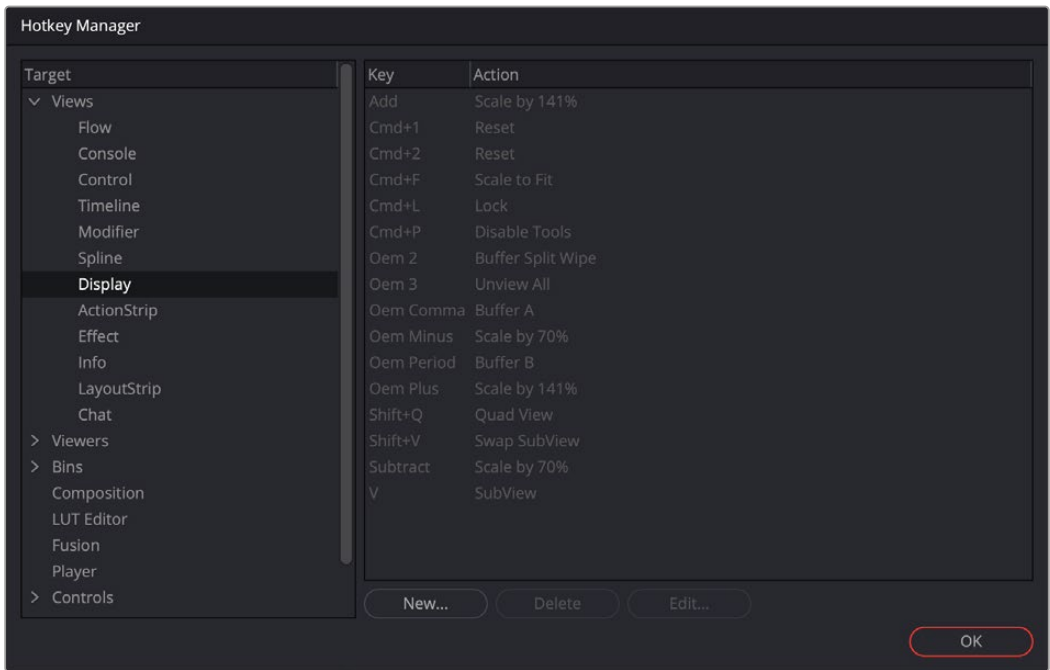
When checked, shot names stored in the EDL file are used to locate the footage.

# Customization

The following section covers the customization of preferences that are not technically part of the Preferences window. Using Fusion Studio's Hotkey Manager window, you can customize the keyboard shortcuts, making the entire process of working in Fusion not only faster but potentially more familiar if you are migrating from another software application. You can also customize Fusion with environment variables to switch between different preferences files, allowing different working setups based on different users or job types. Both of these customization options are only available in Fusion Studio.

## Shortcuts Customization

Keyboard shortcuts can be customized in Fusion Studio. You can access the Hotkey Manager by choosing Customize HotKeys from the View menu.

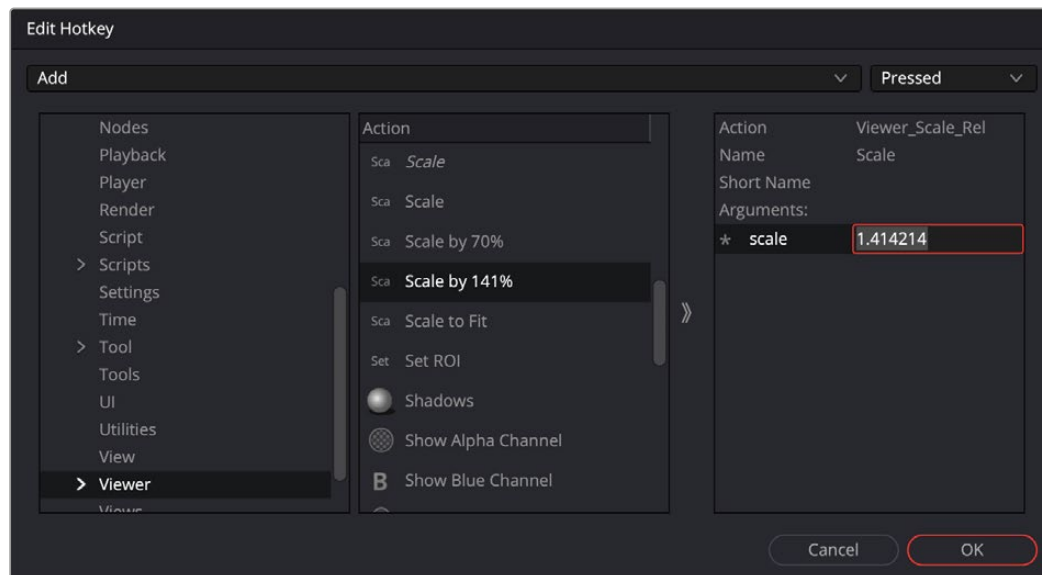


The Hotkey Manager

Fusion has active windows to focus attention on those areas of the interface, like the Node Editor, the viewers, and the Inspector. When selected, a gray border line will outline that section. The shortcuts for those sections will work only if the region is active. For example, Command-F in the

View will scale the image to fit the view area; in the Flow view, Command-F will open the Find tool dialog; and in the Spline editor, it will fit the splines to the window.

On the right is a hierarchy tree of each section of Fusion and a list of currently set hotkeys. By choosing New or Edit, another dialog will appear, which will give specific control over that hotkey.



The Hotkey Editor

Creating a new keyframe will give you the key combo to press, and this Edit Hotkey dialog will appear where the Action can be defined at top right: pressed, repeated, or released. The Name and abbreviated Short Name can be set, as can the Arguments of the action.

## Customizing Preferences

Fusion Studio's preferences configure Fusion's overall application default settings and settings for each new composition. Although you access and set these preferences through the Preferences window, Fusion saves them in a simple text format called Fusion.prefs.

These default preferences are located in a \Profiles\Default folder and shared by all Fusion users on the computer. However, you may want to allow each user to have separate preferences and settings, and this requires saving the preferences to different locations based on a user login.

To change the saved location of the preferences file requires the use of environment variables.

### Setting the Preferences Location

When you first open Fusion, the environment variable FUSION\_PROFILE\_DIR defines the folder that contains the Profiles folder. If this variable defines a valid path, then the preferences are saved to this folder. If the FUSION\_PROFILE\_DIR does not exist, then Fusion attempts to create it. If it cannot create the path, then the preferences are stored in the default Path Map location: AllData\Profiles.

Typically, all users share the same preferences. If you want each user to save separate preferences within their home folder, you must create another environment variable with the name FUSION\_PROFILE (e.g., FUSION\_PROFILE=jane). Using this second environment variable, Fusion will look for the preferences in the PROFILE\_DIR of the user profile. Using a login script, you can make sure the FUSION\_PROFILE is set to the name of the logged in user.



## Creating a Master Preferences File

When working with multiple Fusion users in a studio, you may want to standardize on a few settings. Using the `FUSION_MasterPrefs` environment variable, you can create one or more site-wide preferences in addition to your local personal preferences.

`FUSION_MasterPrefs` must contain the full path to at least one preferences file. If you have multiple preferences paths, separate them using semicolons. Fusion does not write to these prefs files, and they may contain a subset of all available settings. You may change settings in these files and use them only where local prefs do not already exist unless you set the Locked flag.

## Locking Preferences

If the line “Locked = true,” appears in the main table of a master file, all settings in that file are locked and override any other preferences. Locked preferences cannot be altered by the user.